

SAIR

Spatial AI & Robotics Lab

CSE 473/573

L13: STEREO VISION

Chen Wang

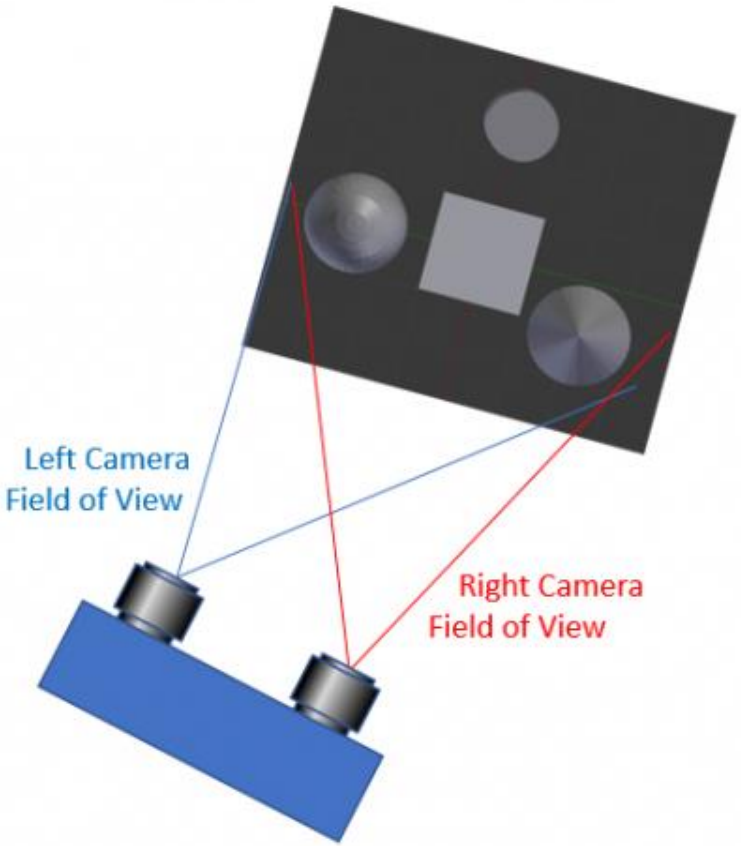
Spatial AI & Robotics Lab

Department of Computer Science and Engineering

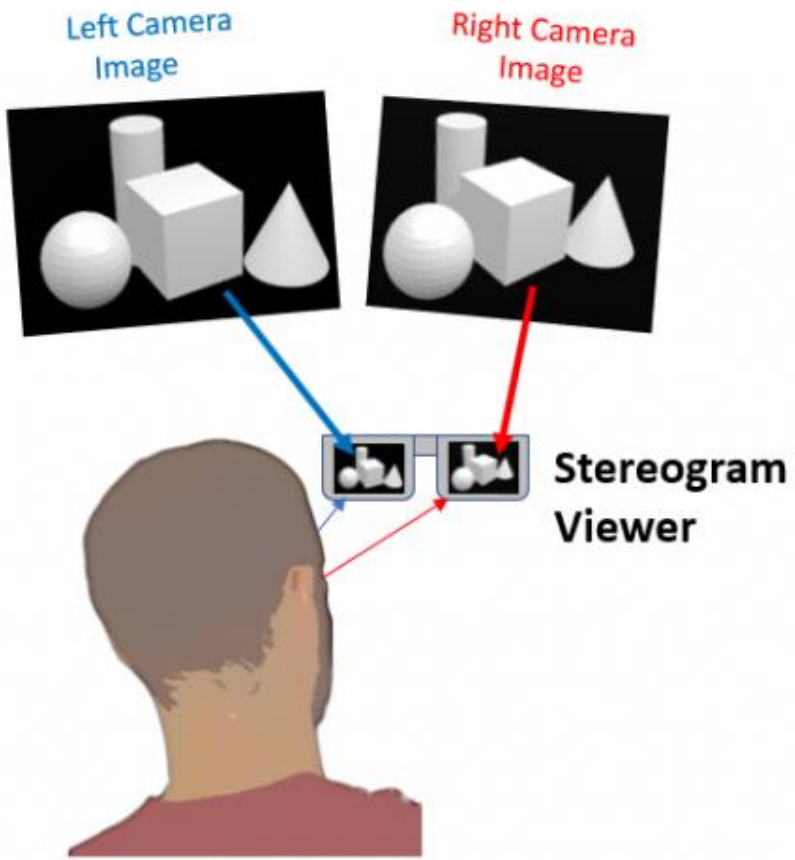
University at Buffalo The State University of New York

Many Slides from Lana Lazebnik

Stereo Vision



Stereo Camera



Why multiple views?

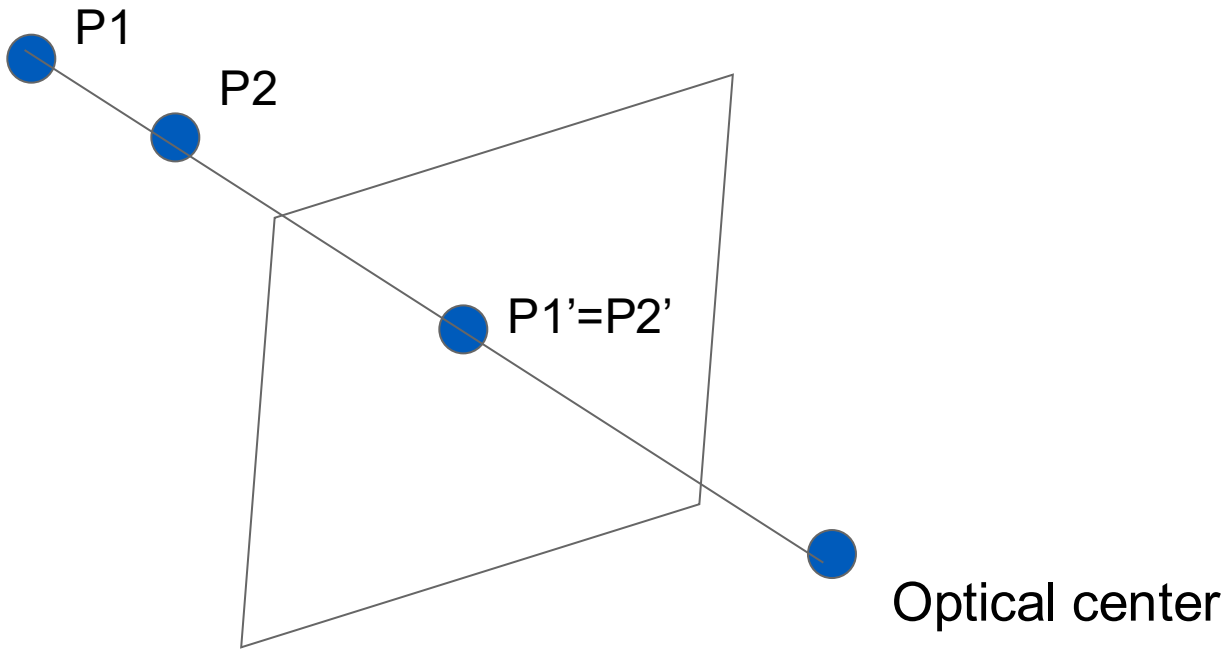
- Structure and depth are inherently ambiguous from single views.





Why multiple views?

- Structure and depth are inherently ambiguous from single views.



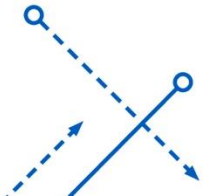
Stereo photography and stereo viewers

- Invented by Sir Charles Wheatstone, 1838
- Take two pictures of the same subject from two slightly different viewpoints and display:
  - Each eye sees only one of the images



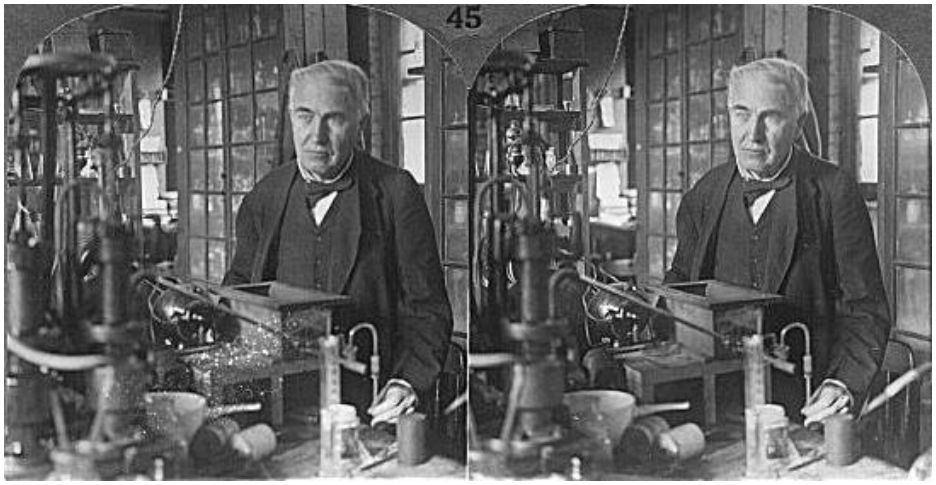


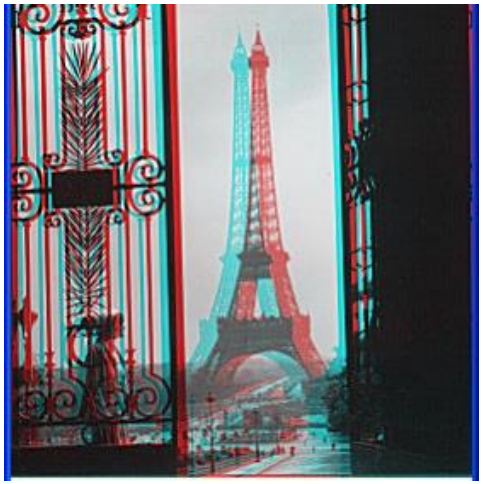


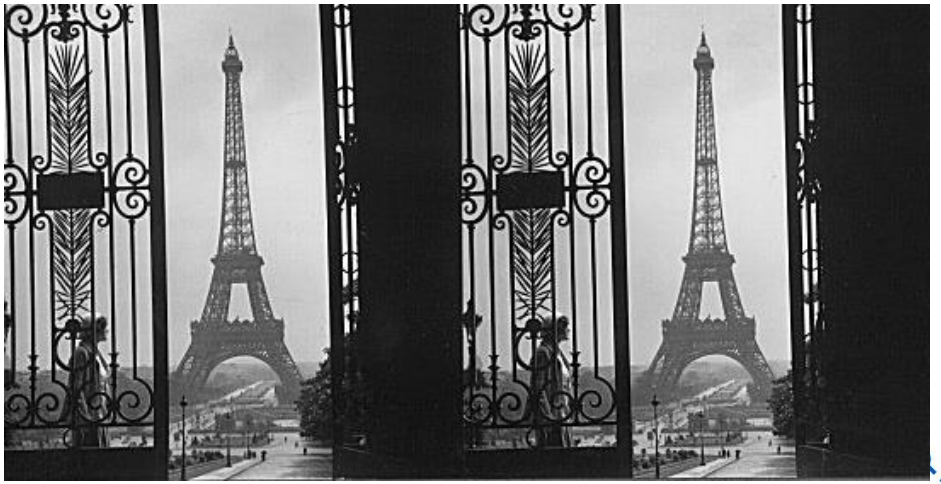


Stereo photography and stereo viewers





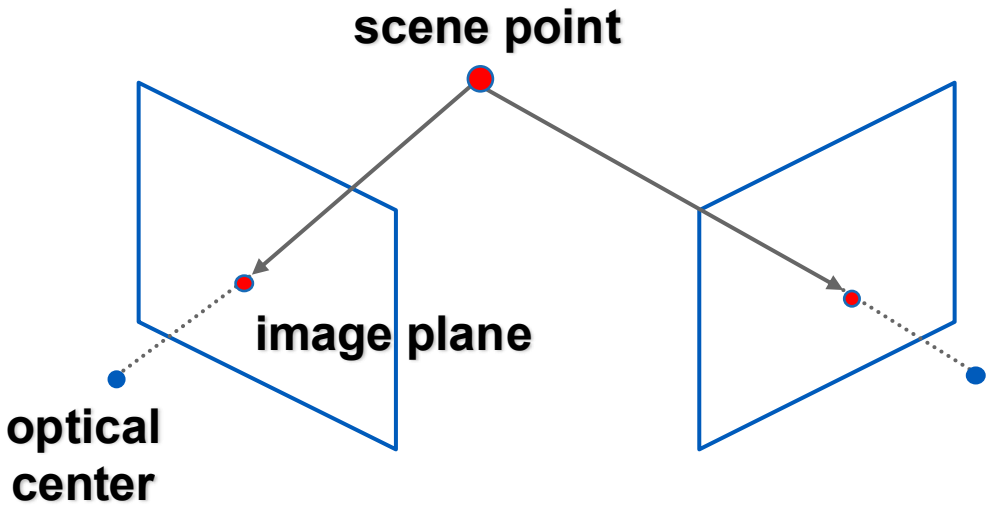


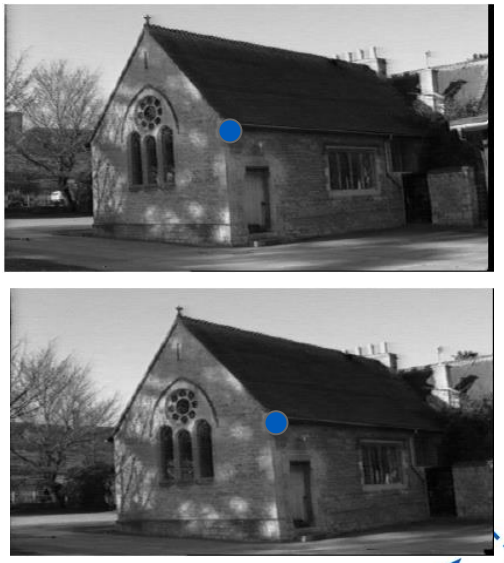




Estimating depth with stereo

- We need to know:
  - Info on Camera Pose
    - **Calibration**
  - Image Point Correspondences
    - **Feature Detection & Matching**





Stereo Vision

- Structure from motion (SFM)
  - Shape from “motion” between two views
  - Infer **3D shape** of scene from two (multiple) views.



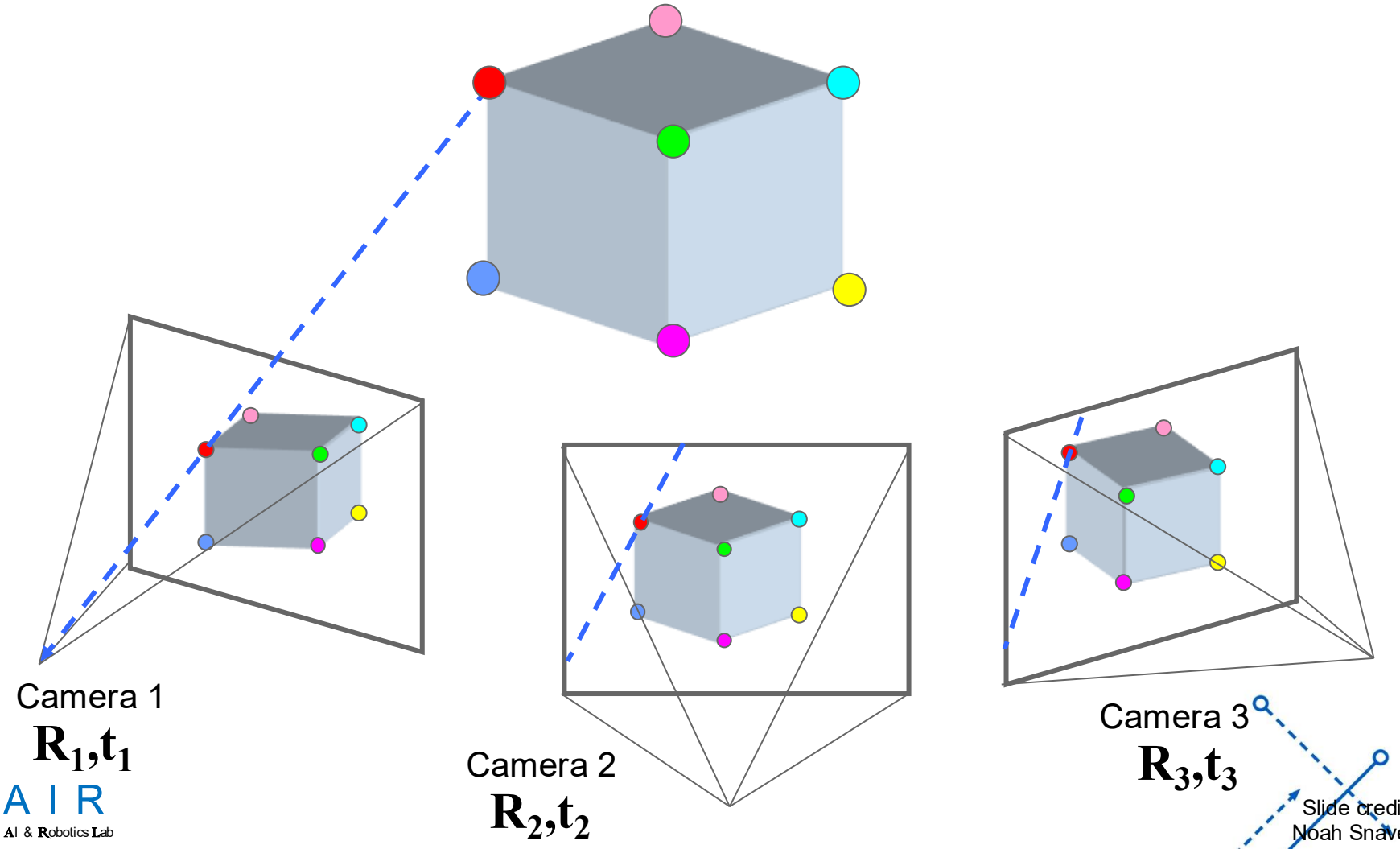
Two cameras, simultaneous views



Single moving camera and static scene

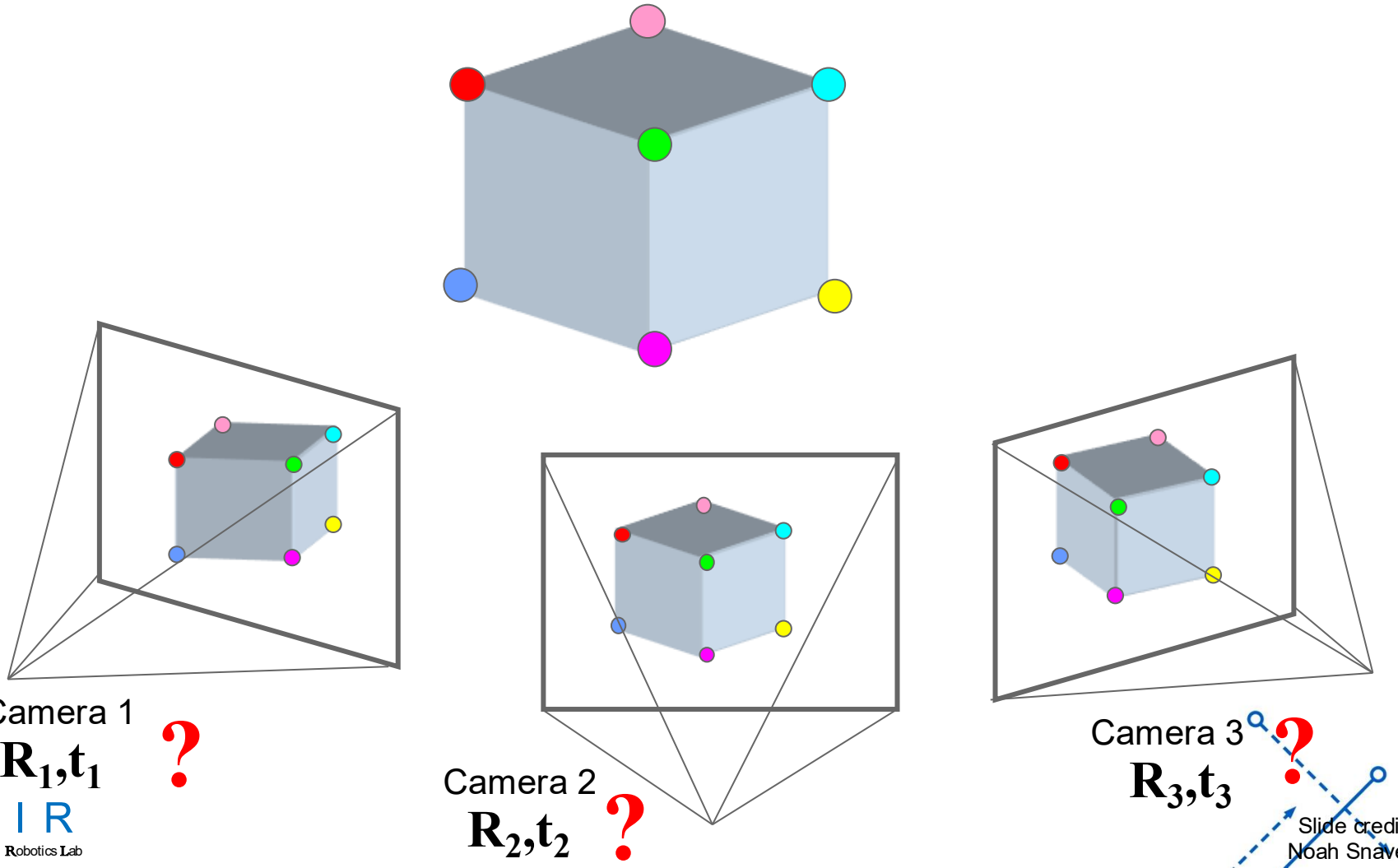
Structure from motion (SFM)

- **Stereo correspondence:** Given a point in one of the images, where could its corresponding points be in the other images?



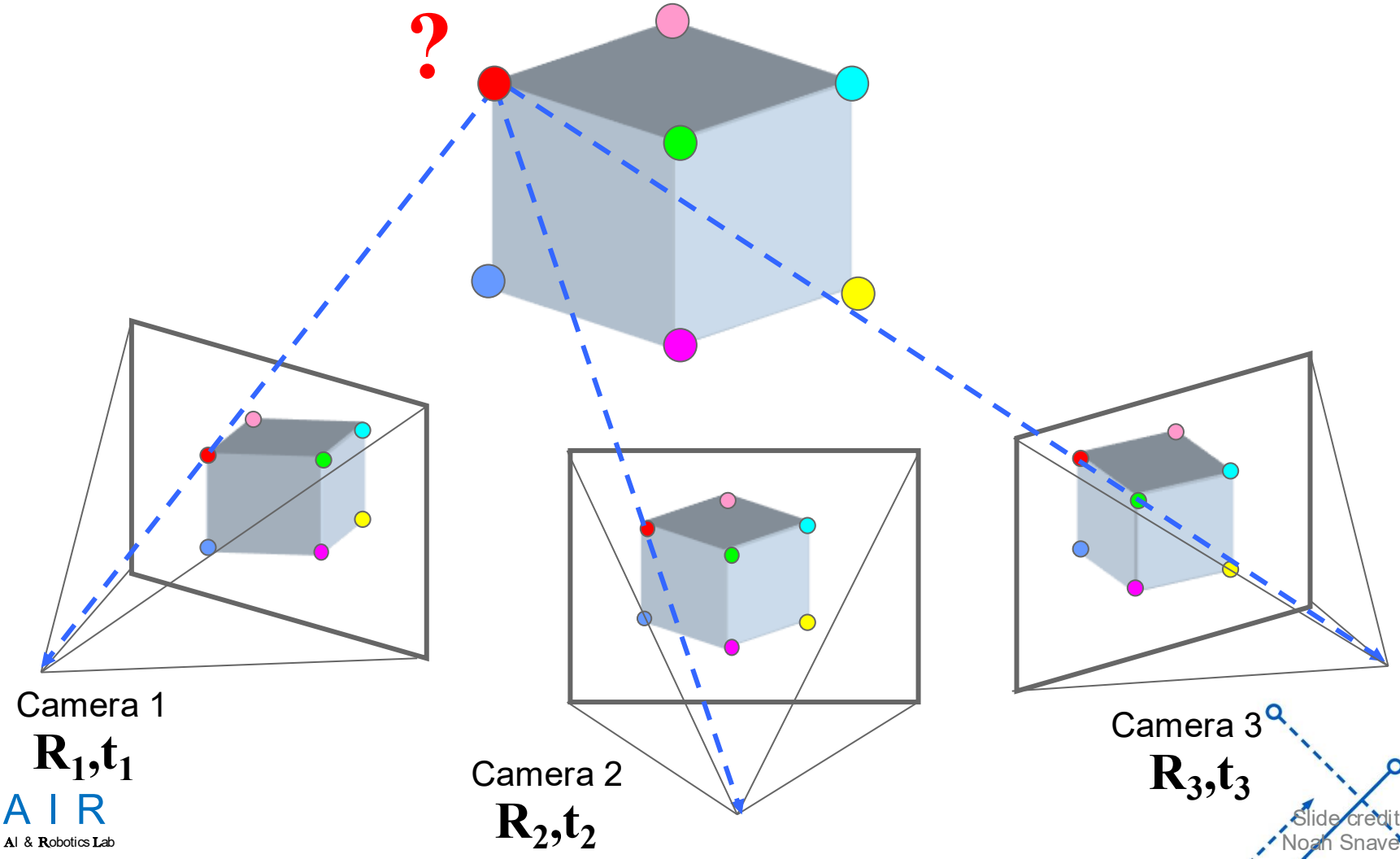
Structure from motion (SFM)

- **Motion:** Given a set of corresponding points in two or more images, compute the camera motion parameters



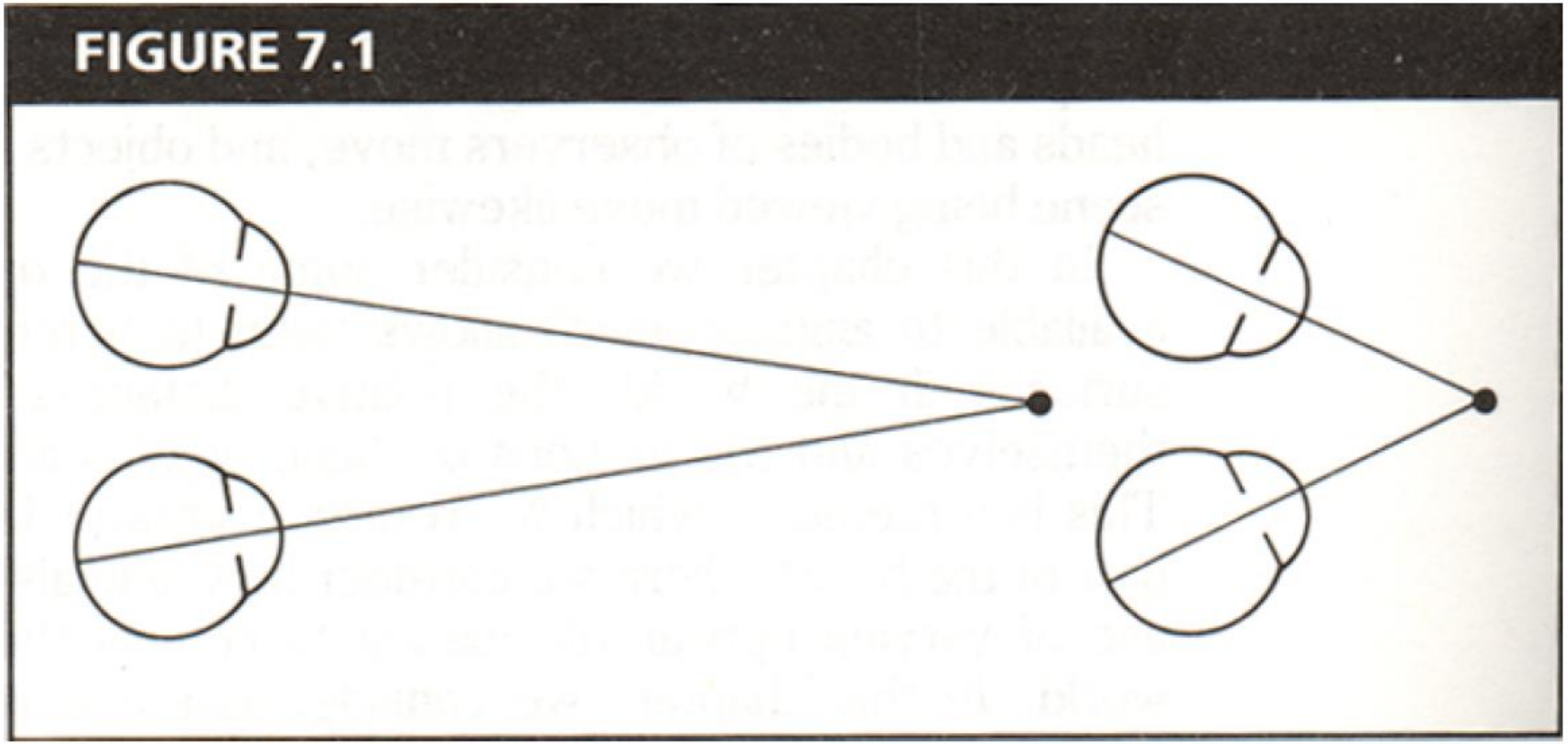
Structure from motion (SFM)

- **Structure:** Given projections of the same 3D point in two or more images, compute the **3D coordinates of that point**.



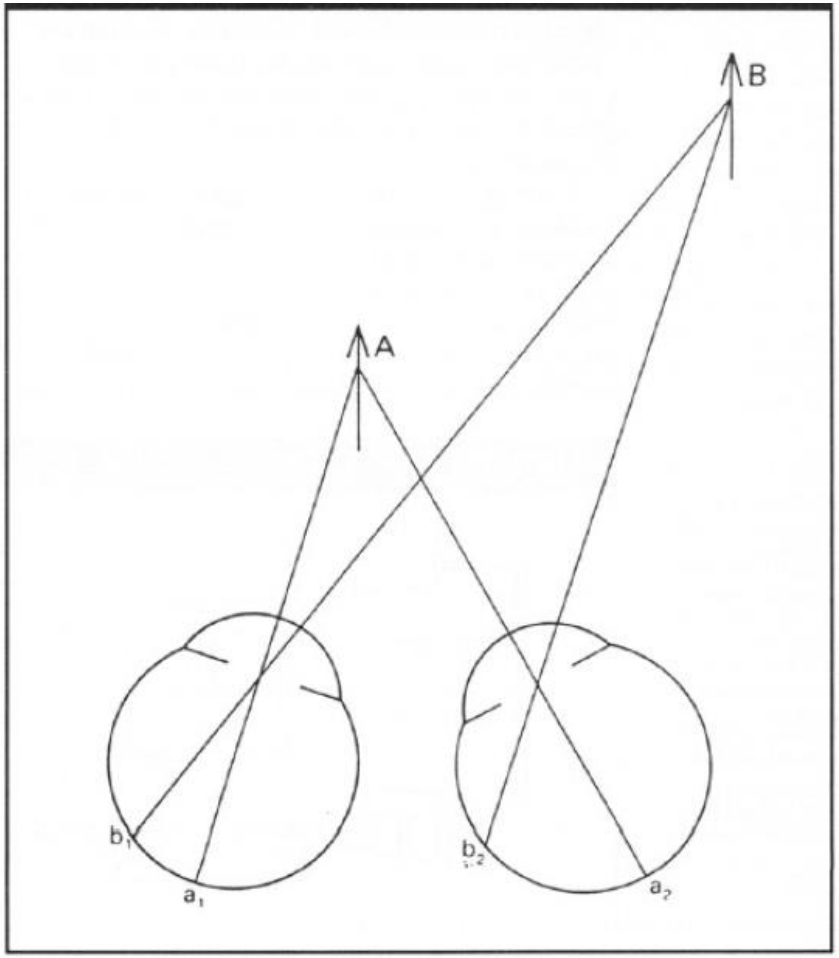
Human stereopsis: disparity

- Human eyes **fixate** on point in space – rotate so that corresponding images form in centers of fovea.



Disparity

- **Disparity** occurs when eyes fixate on one object; others appear at different visual angles

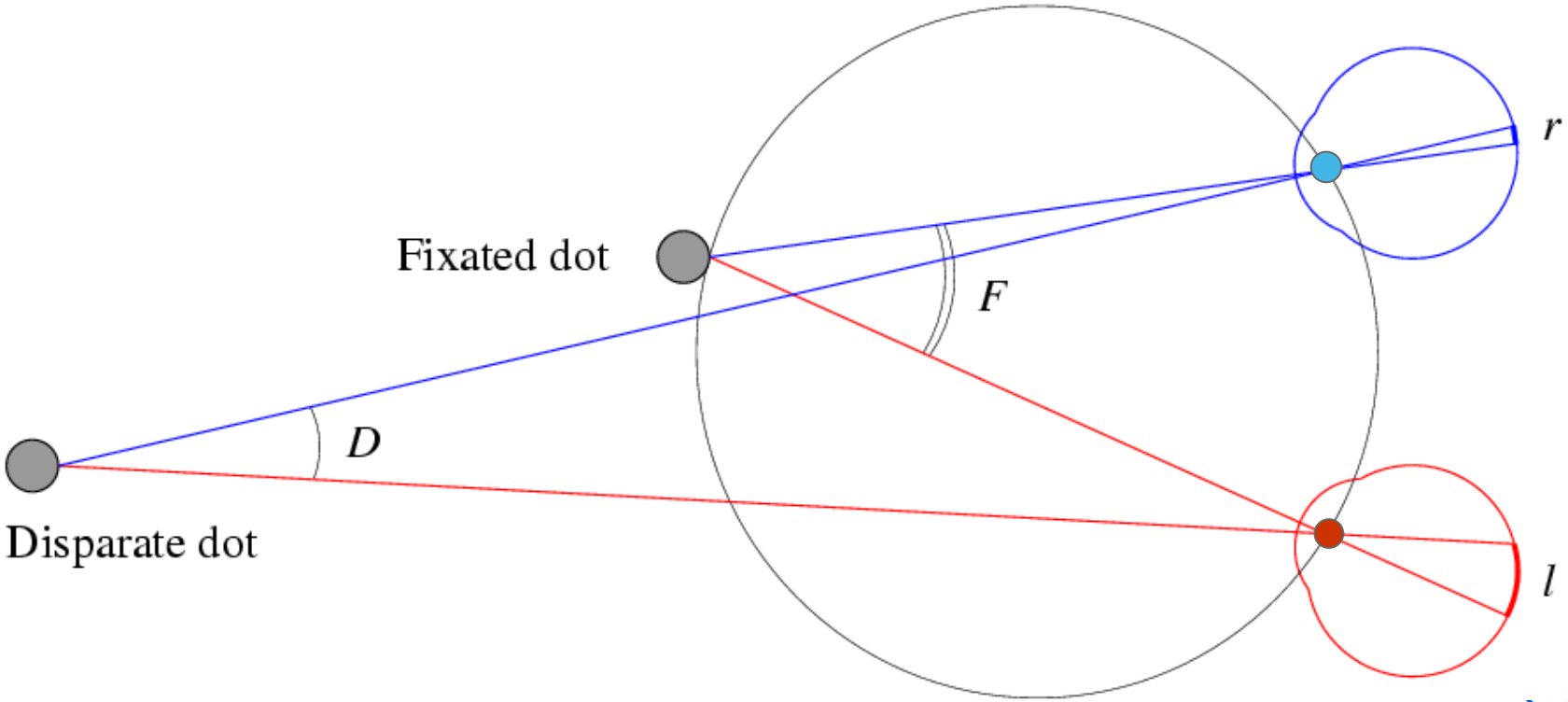


Disparity

- Disparity:

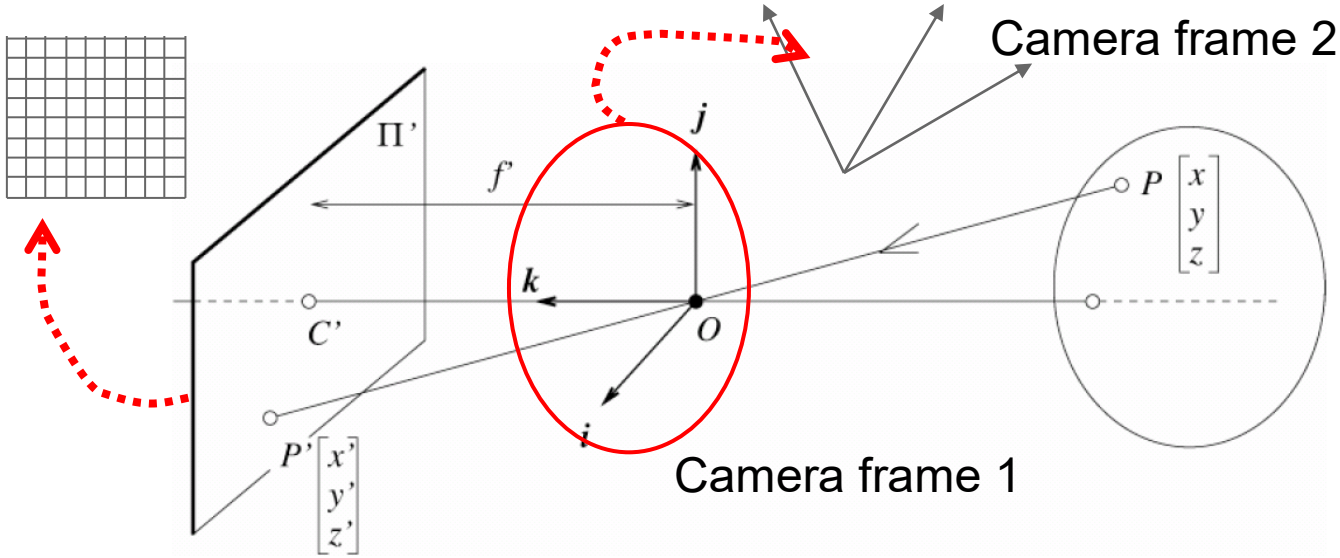
$$d = r - l = D - F.$$

$$\text{radian} = \frac{\text{arc length}}{\text{radius}}$$



Camera parameters (Recap)

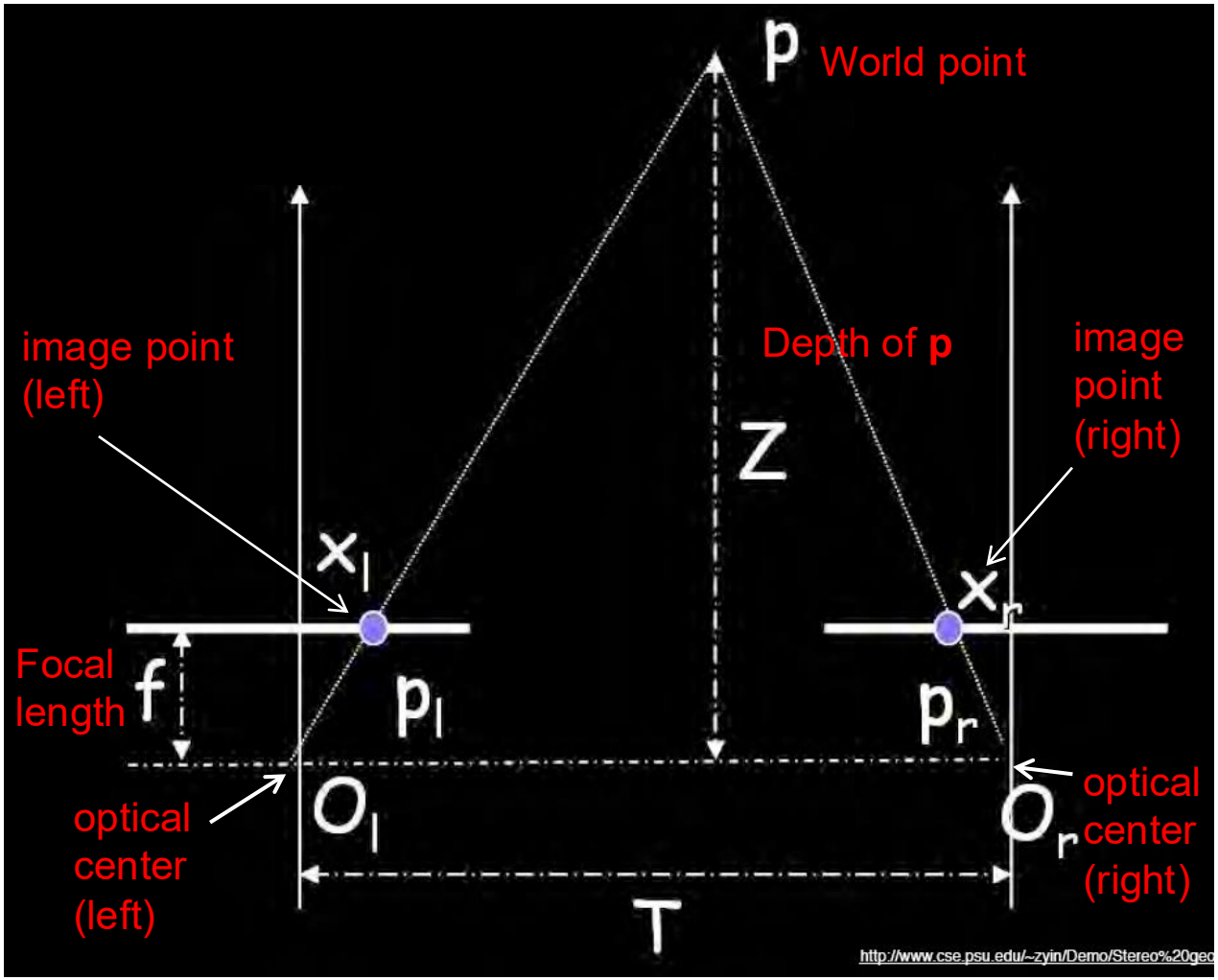
- *Intrinsic* params:
  - Focal length, image center, and (radial distortion parameters).
  - Coordinates relative to camera ←→ Pixel coordinates
- *Extrinsic* params:
  - Rotation matrix and translation vector
  - Camera frame 1 ←→ Camera frame 2



We'll assume for now that these parameters are given and fixed.

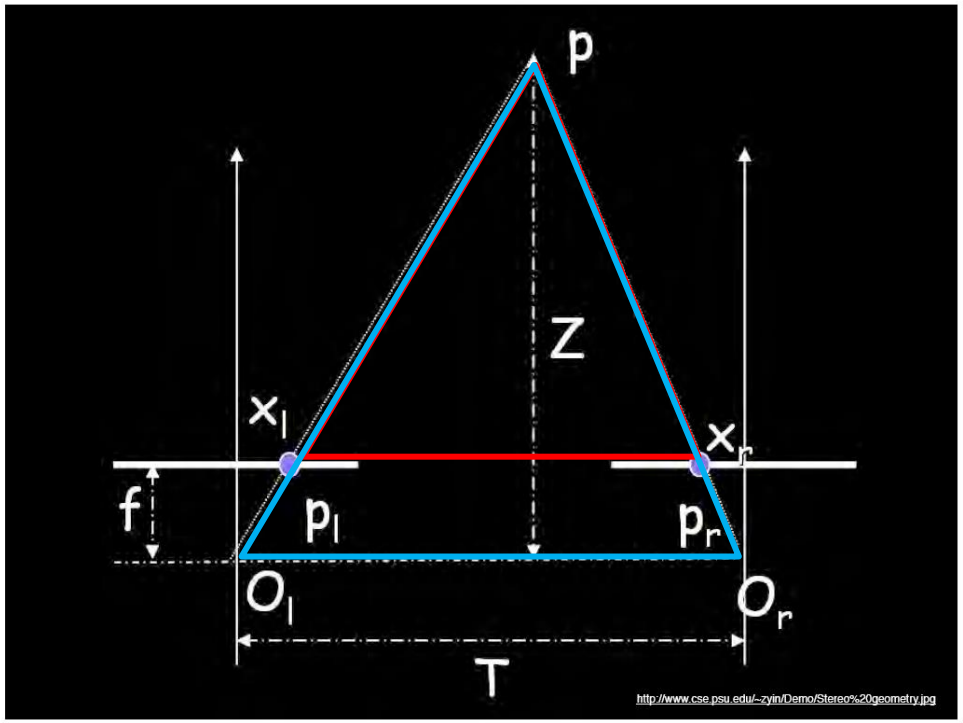
Geometry for a simple stereo system

- Assuming parallel optical axes, known camera parameters (i.e., calibrated cameras):



Geometry for a simple stereo system

- Assume parallel optical axes, known camera parameters, i.e., calibrated cameras. **What is expression for Z?**
- Similar triangles (p_l, P, p_r) and (O_l, P, O_r):



$$\frac{T - x_l + x_r}{Z - f} = \frac{T}{Z}$$

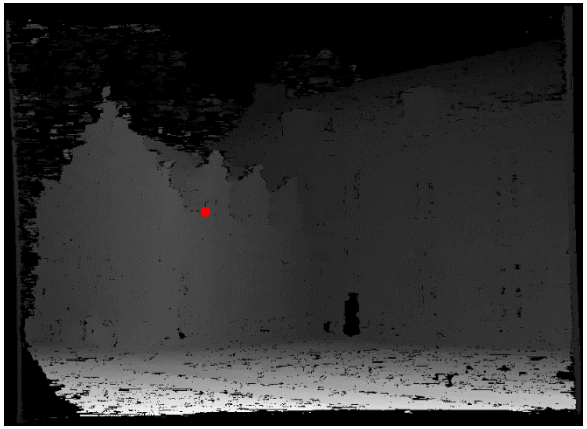
$$Z = f\frac{T}{x_l - x_r}$$

Disparity

Depth from disparity

- If we could find the **corresponding points** in two images, we could **estimate relative depth.**

Image $I(x, y)$ | Disparity map $D(x, y)$ | Image $I'(x', y')$


$$(x', y') = (x + D(x, y), y)$$

Depth from disparity

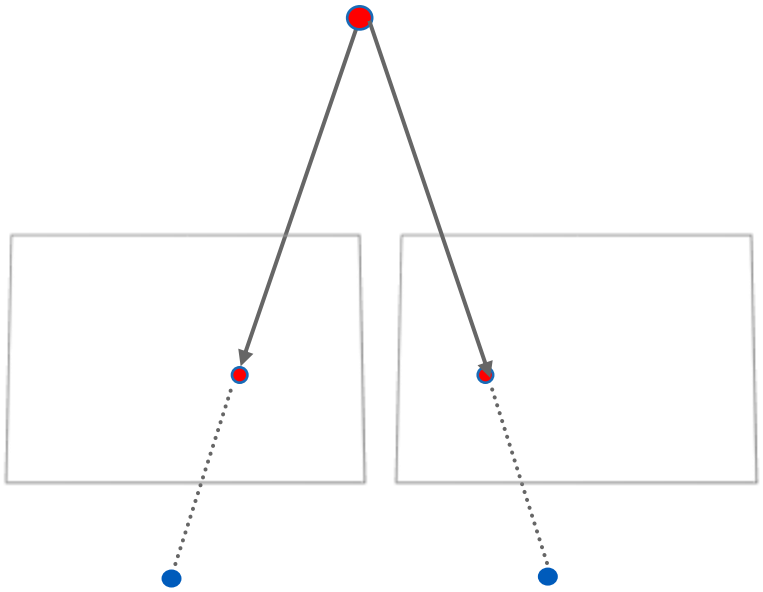
- Where do we need to search?



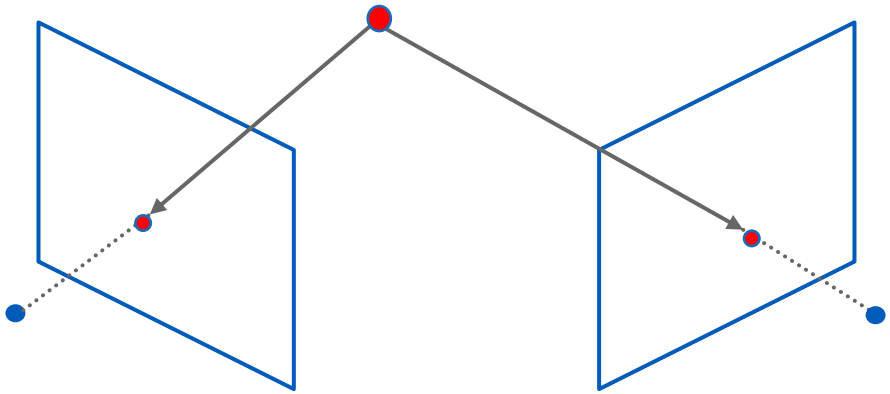


General case (calibrated cameras)

- The two cameras need not have parallel optical axes.



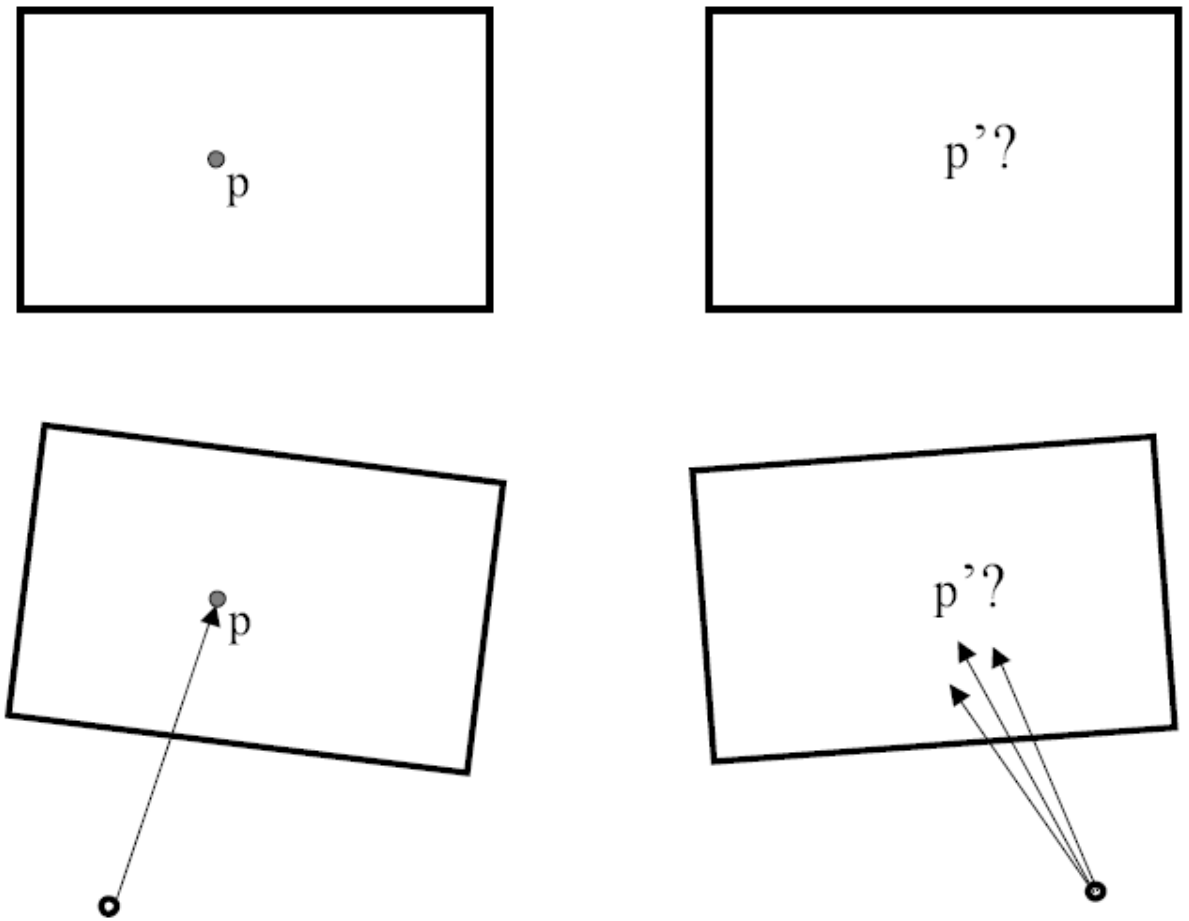
Vs.





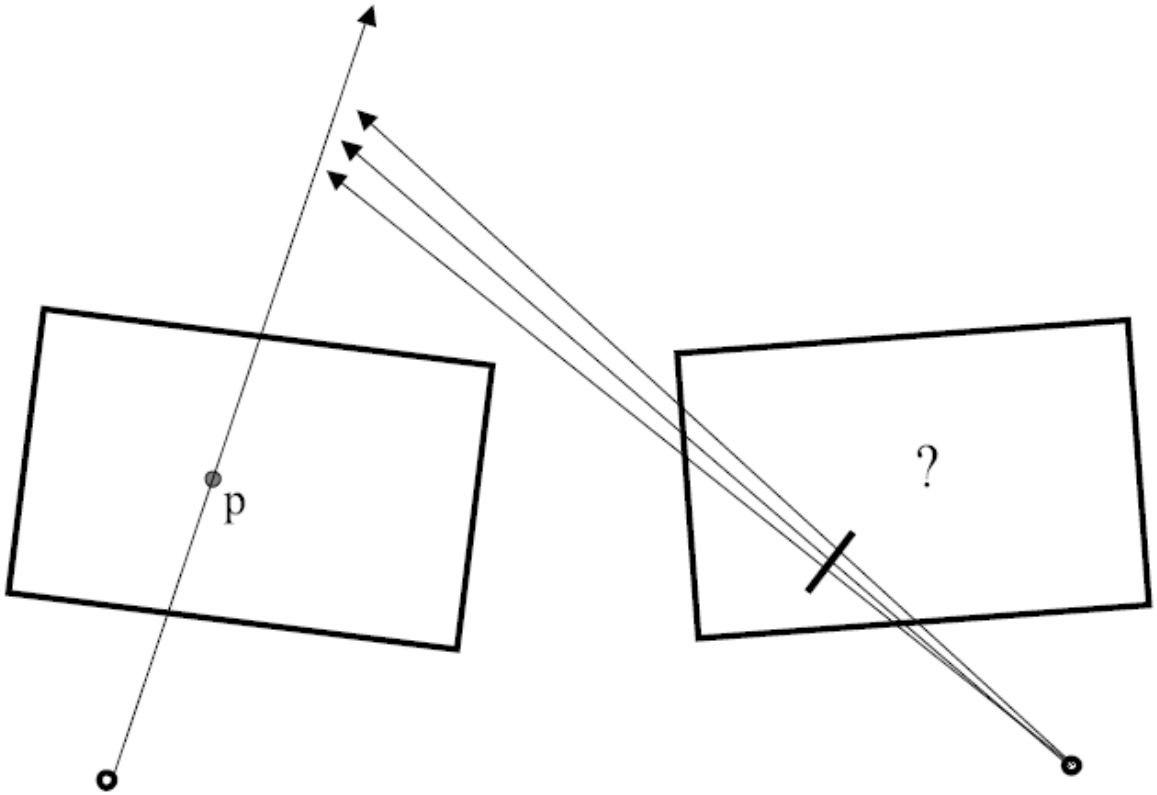
Stereo correspondence constraints

- Given p in left, where can corresponding point p' be?



Stereo correspondence constraints

- Given p in left, where can corresponding point p' be?



Correspondence problem

- Multiple match hypotheses satisfy epipolar constraint, but which is correct?

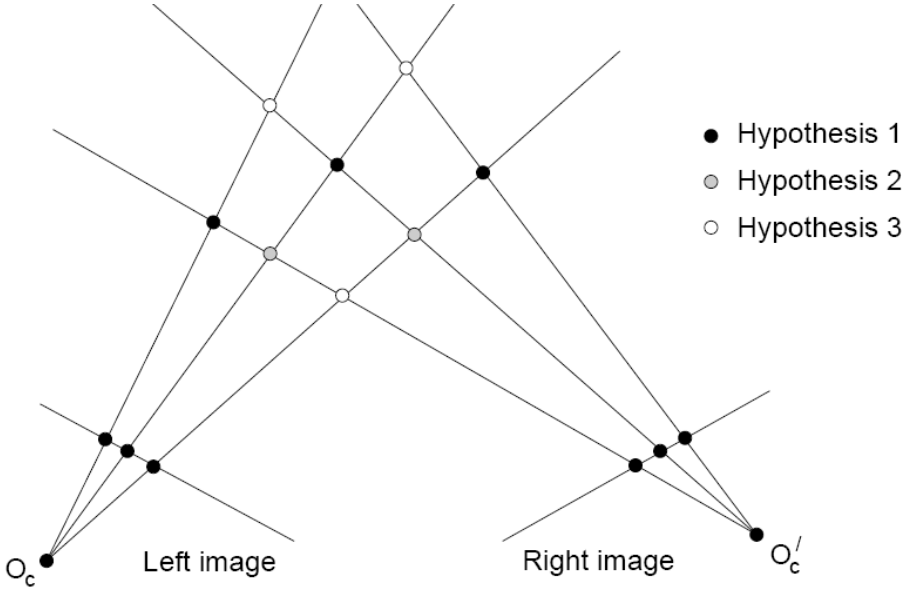


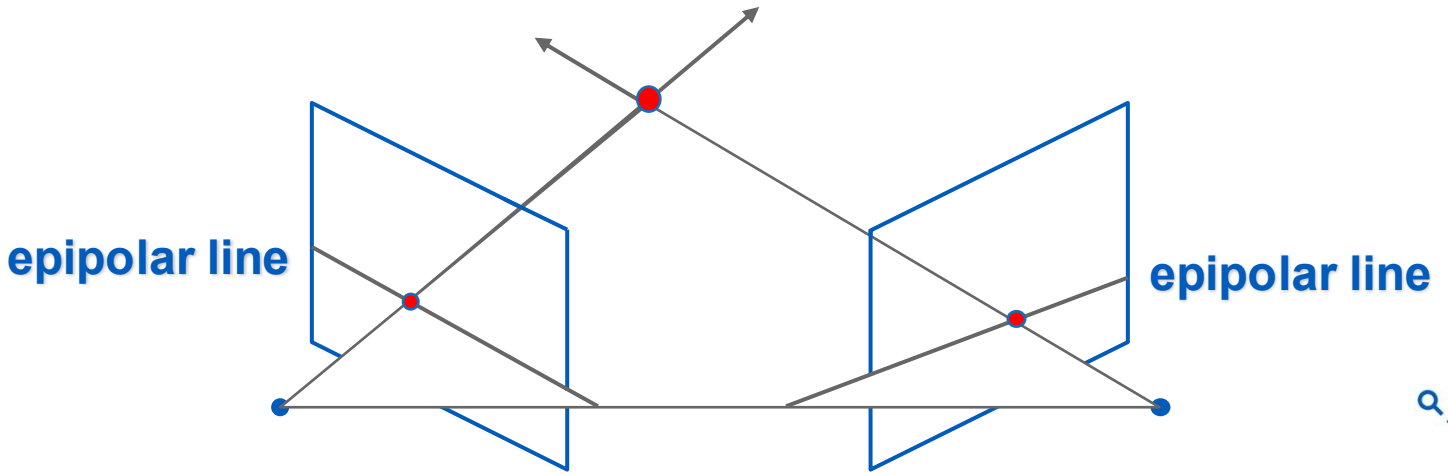




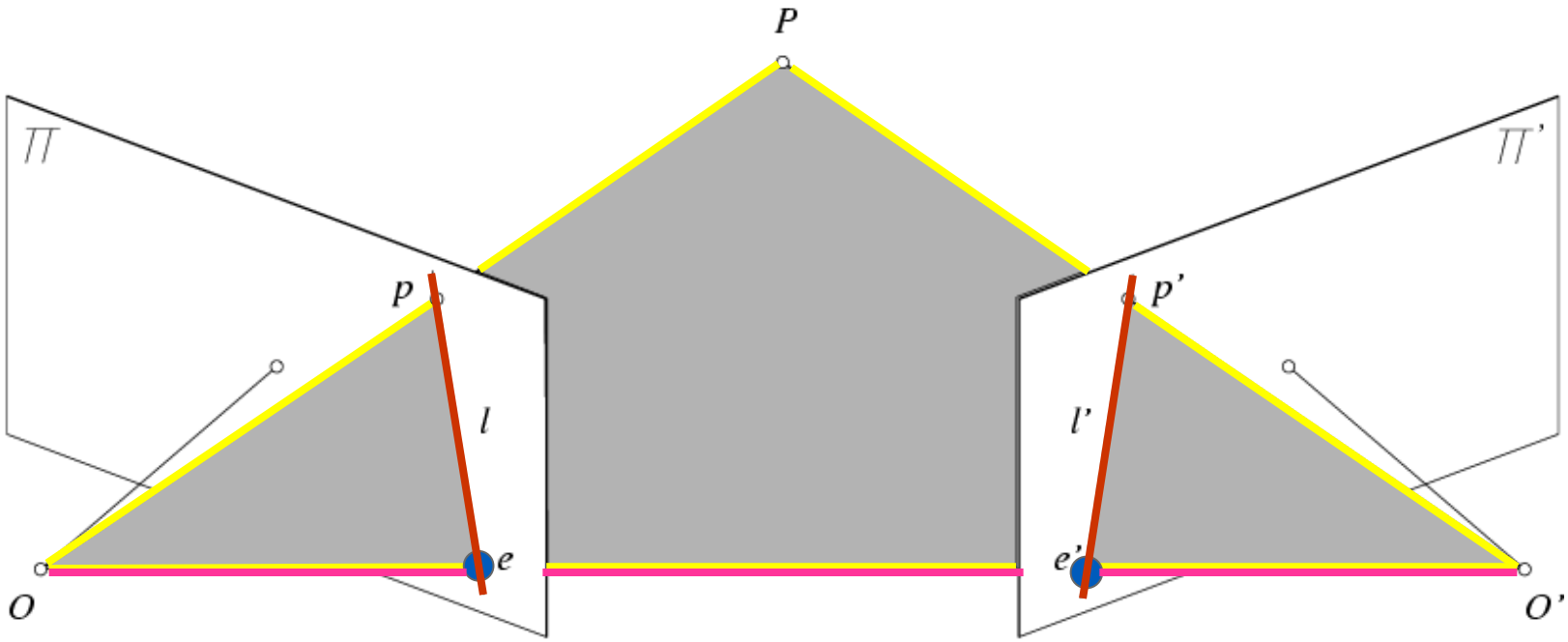
Figure from Gee & Cipolla 1999

Stereo correspondence constraints

- Geometry of two views allows us to constrain where the corresponding pixel for image points in the first view must occur in the second view.
- Epipolar constraint
  - Reduces correspondence problem to 1D search along conjugate epipolar lines



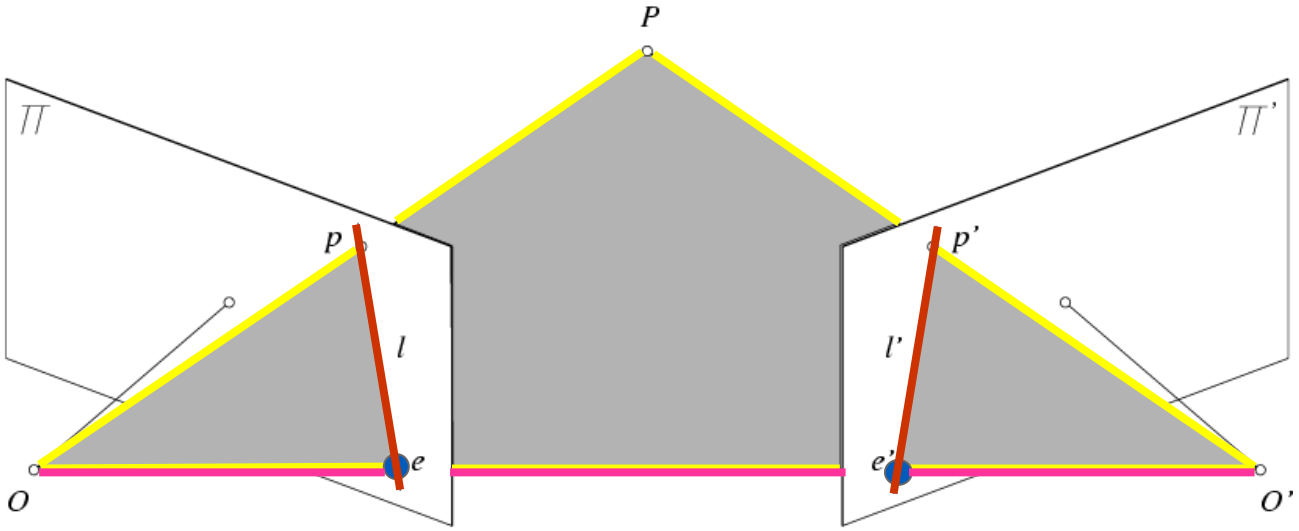
Epipolar geometry



- Epipolar Plane
- Baseline
- Epipoles
- Epipolar Lines

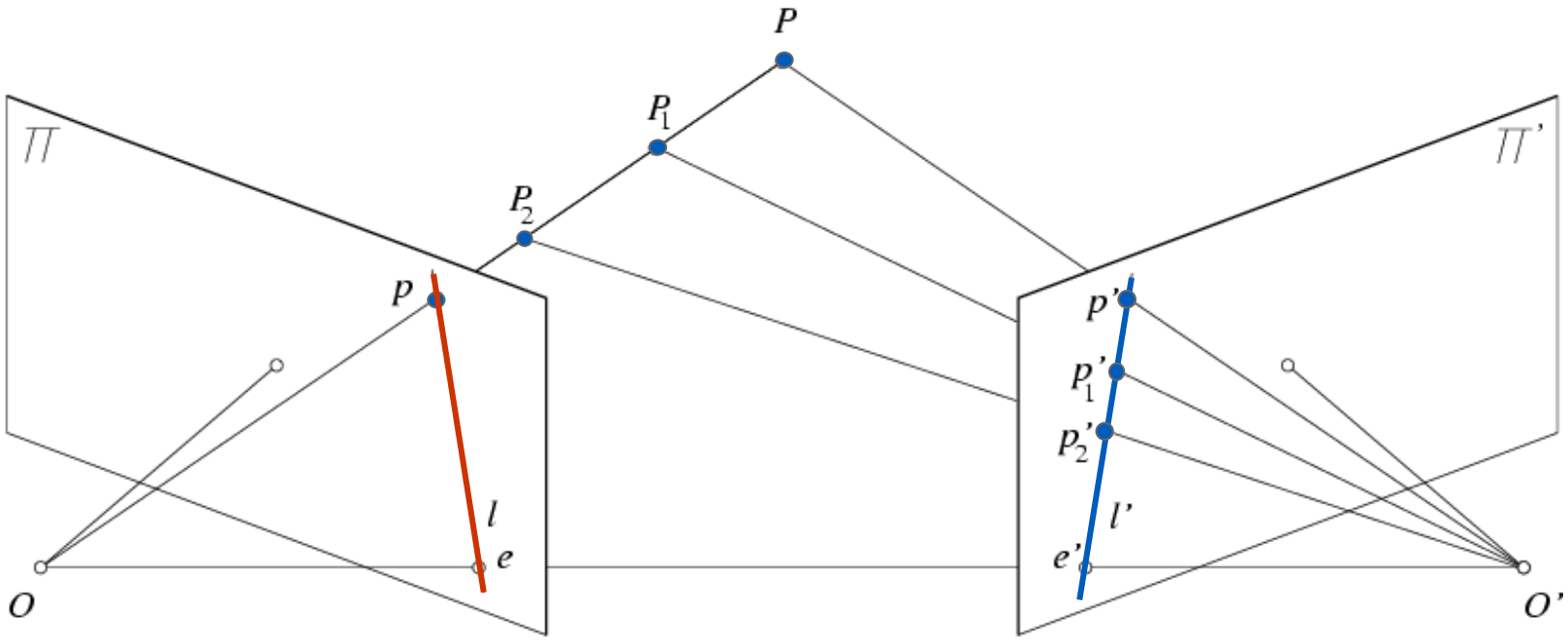
Epipolar geometry: terms

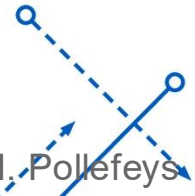
- **Baseline**: line joining the camera centers
- **Epipole**: point of intersection of baseline with the image plane
- **Epipolar plane**: plane containing baseline and world point
- **Epipolar line**: intersection of epipolar plane with image plane
- An epipolar plane intersects the left and right image planes in epipolar lines
- All epipolar lines intersect at the epipole



Epipolar constraint

- Potential matches for p must lie on epipolar line l'.
- Potential matches for p' must lie on epipolar line l.

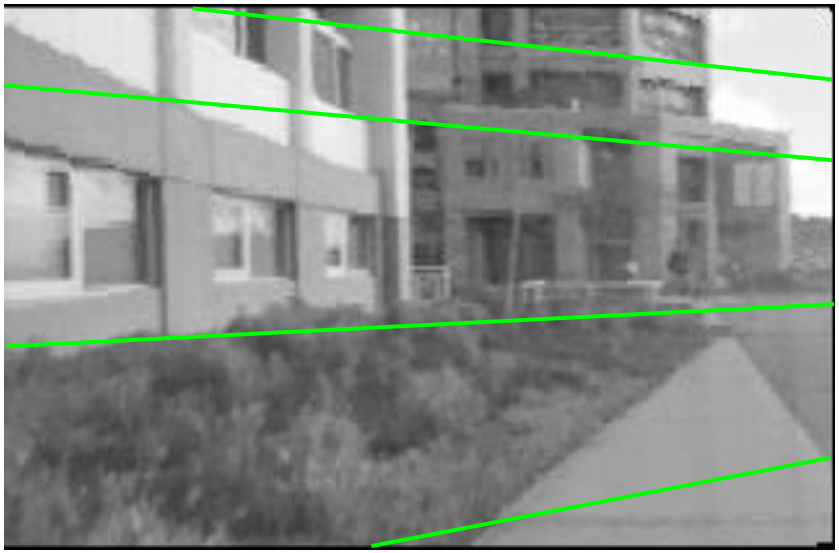




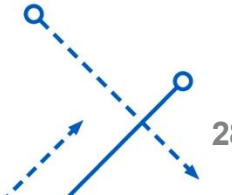
Source: M. Pollefeys

Rectification

- Searching along epipolar lines at arbitrary orientation is intuitively expensive.
- We prefer searching along the image row.
  - Epipolar lines parallel to the rows of the image.
- This transformation is called *rectification*.

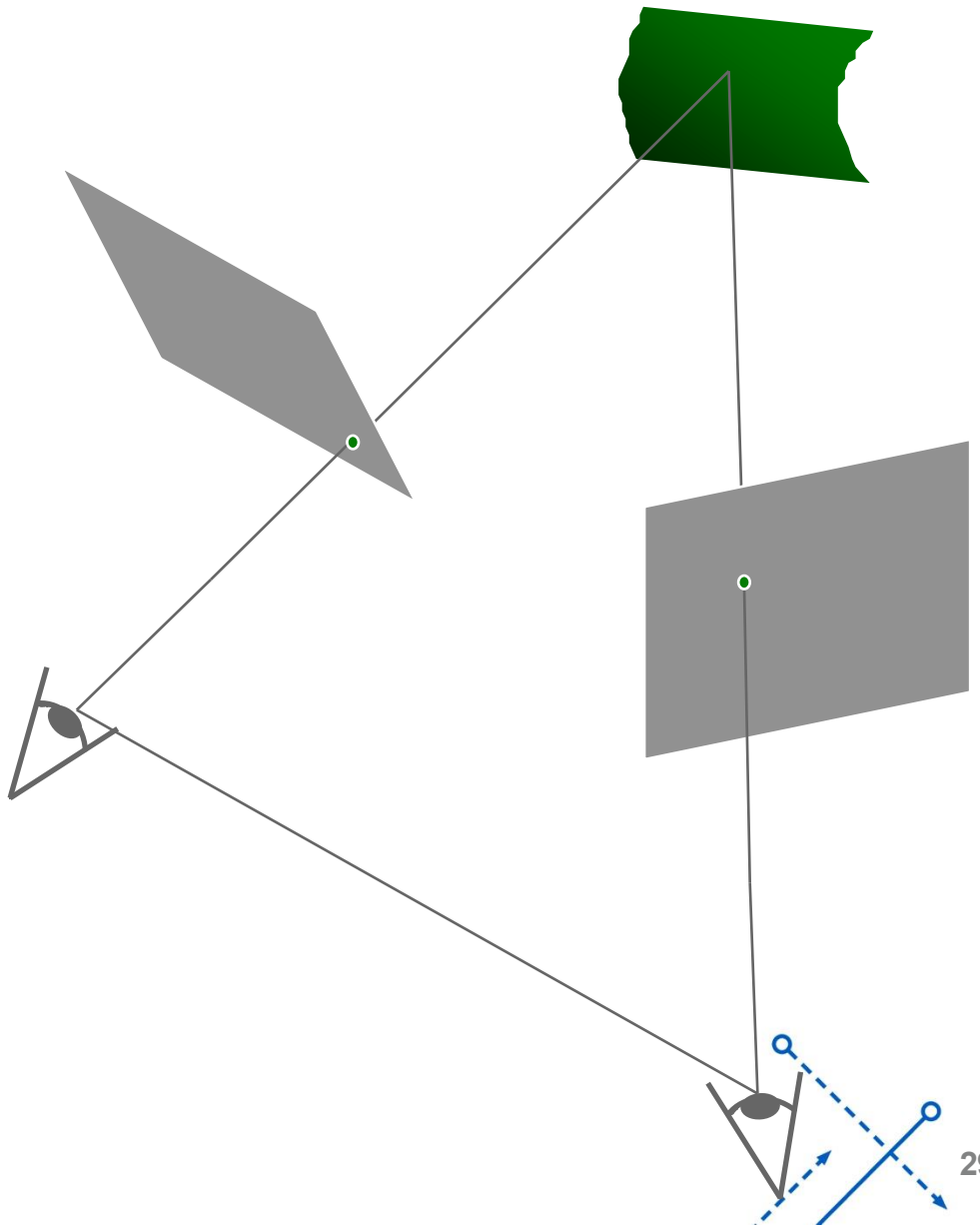






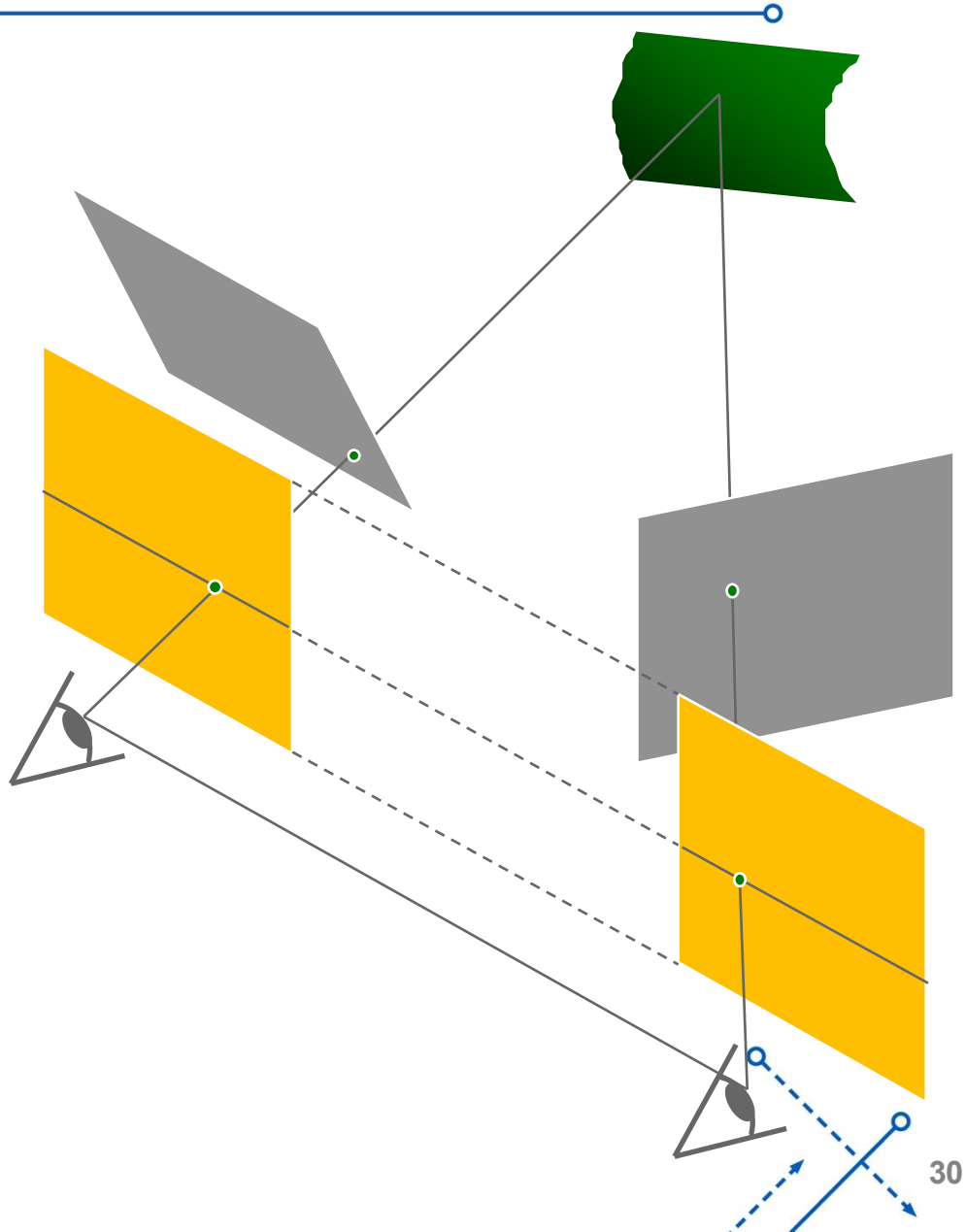
Stereo Image Rectification

- Reproject image planes onto a common plane parallel to the line between optical centers

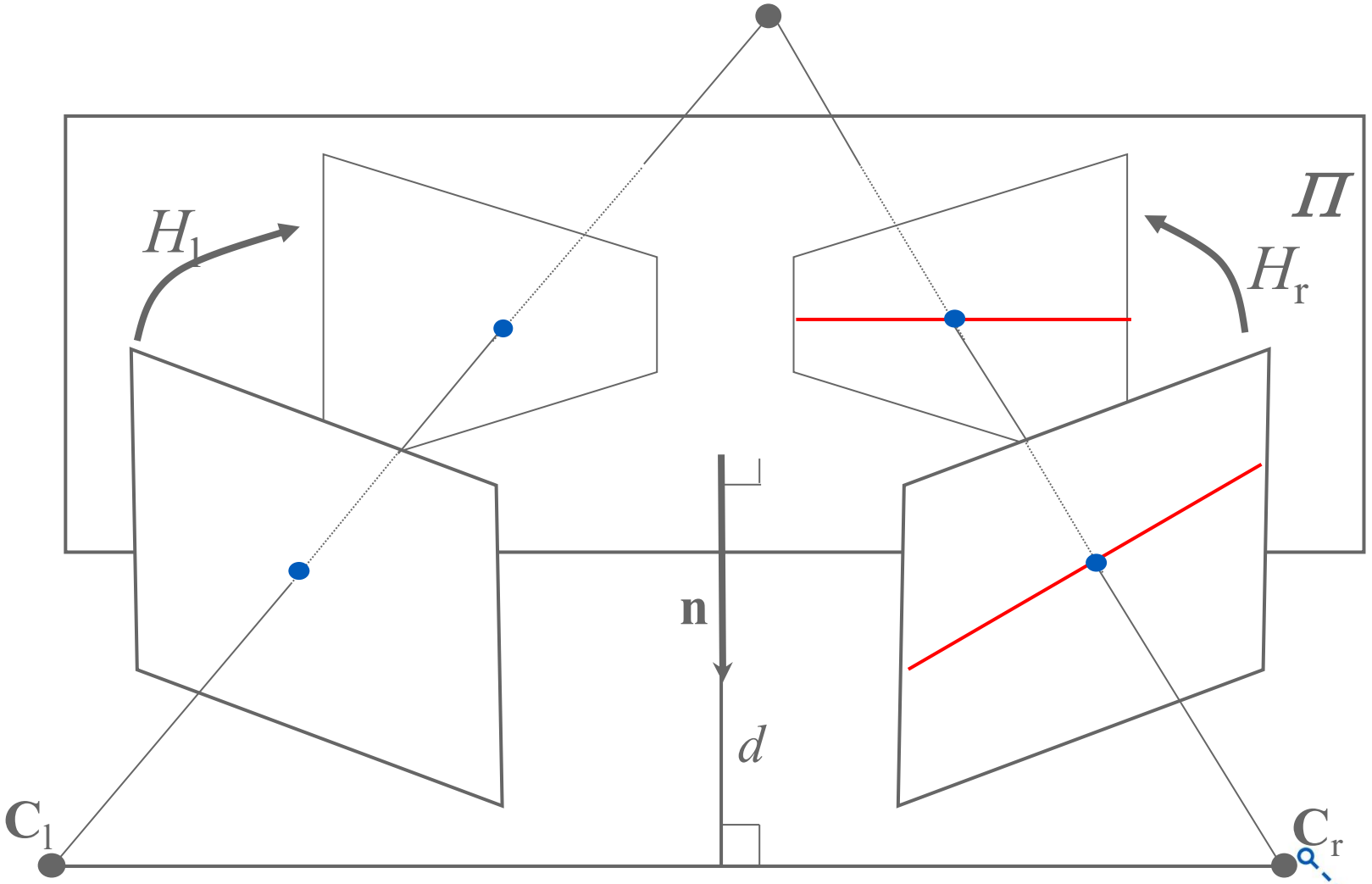


Stereo Image Rectification

- Reproject image planes onto a common plane parallel to the line between optical centers
- pixel motion is horizontal after this transformation



Stereo Image Rectification

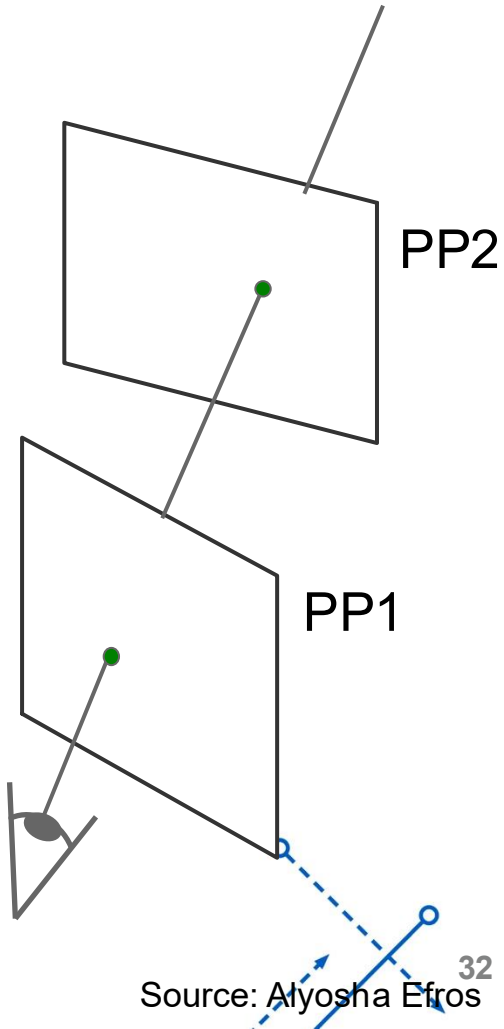


Rectification using Homography (Recap)

- How to relate two images from the same camera center?
  - How to map a pixel from PP1 to PP2?
- Take as a 2D **image warp** using projective transform**.**
- A **projective transform** is a mapping between any two PPs with the same center of projection
  - rectangle should map to arbitrary quadrilateral
  - parallel lines aren't preserved.
  - but straight lines are preserved.
- Called **Homography**
- **Refer to L9: Alignment**

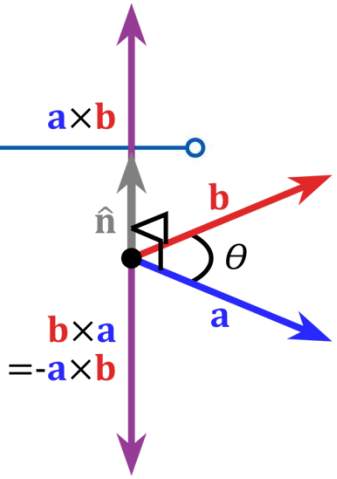
$$\begin{bmatrix} wx' \\ wy' \\ w \end{bmatrix} = \begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$\mathbf{p'}$ $\quad$ $\mathbf{H}$ $\quad$ $\mathbf{p}$



Source: Alyosha Efros

Cross product (Review)



- Cross product of two 3D vector

$$\mathrm{a} \times b = \hat{n}\,|a|\,|b|\sin\theta$$

$$\begin{aligned}\mathbf{a}\times\mathbf{b} &= (a_1\mathbf{i}+a_2\mathbf{j}+a_3\mathbf{k})\times(b_1\mathbf{i}+b_2\mathbf{j}+b_3\mathbf{k})\\ &= a_1b_1(\mathbf{i}\times\mathbf{i})+a_1b_2(\mathbf{i}\times\mathbf{j})+a_1b_3(\mathbf{i}\times\mathbf{k})+\\ &\quad a_2b_1(\mathbf{j}\times\mathbf{i})+a_2b_2(\mathbf{j}\times\mathbf{j})+a_2b_3(\mathbf{j}\times\mathbf{k})+\\ &\quad a_3b_1(\mathbf{k}\times\mathbf{i})+a_3b_2(\mathbf{k}\times\mathbf{j})+a_3b_3(\mathbf{k}\times\mathbf{k})\end{aligned}$$

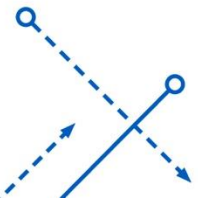
$$\mathbf{a}\times\mathbf{b} = \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\ a_1 & a_2 & a_3\\ b_1 & b_2 & b_3\end{vmatrix}$$

- Skew-symmetric Matrix

$$\mathbf{a}\times\mathbf{b} = [\mathbf{a}]_\times\mathbf{b} = \begin{bmatrix}0 & -a_3 & a_2\\ a_3 & 0 & -a_1\\ -a_2 & a_1 & 0\end{bmatrix}\begin{bmatrix}b_1\\ b_2\\ b_3\end{bmatrix} \qquad [\mathbf{a}]_\times = \begin{bmatrix}0 & -a_3 & a_2\\ a_3 & 0 & -a_1\\ -a_2 & a_1 & 0\end{bmatrix}$$

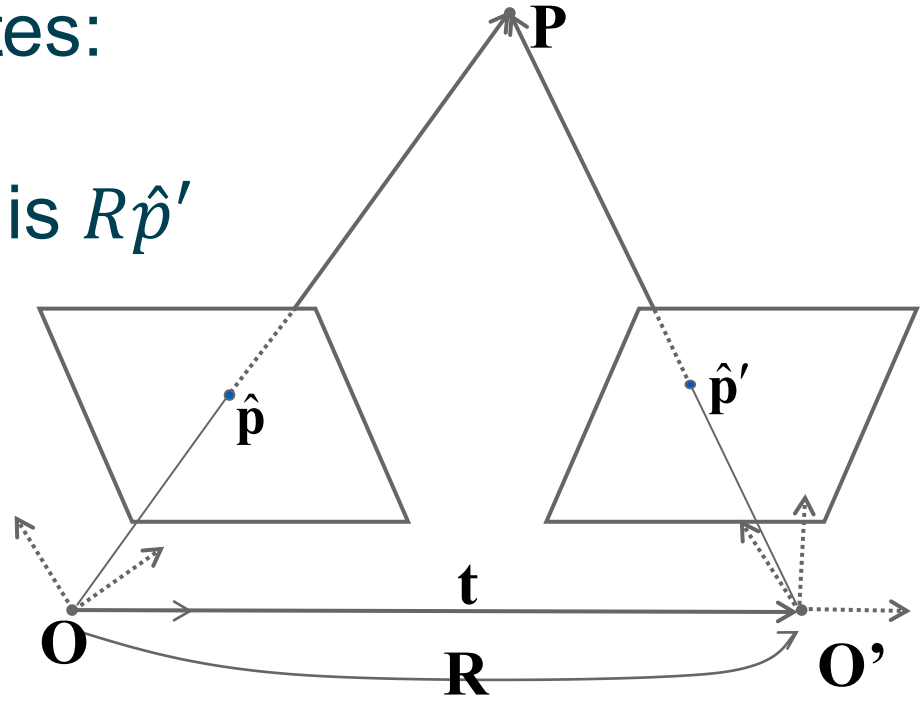
- Properties:

$$\mathrm{a}^{\mathrm{T}}(a\times b) = b^T(a\times b) = 0$$



Essential Matrix

- Due to cross product properties
  - $\overrightarrow{OP} \cdot \left(\overrightarrow{OO'} \times \overrightarrow{O'P}\right) = 0$
- In homogeneous coordinates:
  - Transform O to align O'
  - Then direction of $\hat{p}'$ in O is $R\hat{p}'$
  - $\hat{\boldsymbol{p}}^T(\boldsymbol{t} \times R\hat{\boldsymbol{p}}') = 0$
  - $\hat{\boldsymbol{p}}^T([\boldsymbol{t}]_\times R)\hat{\boldsymbol{p}}' = 0$

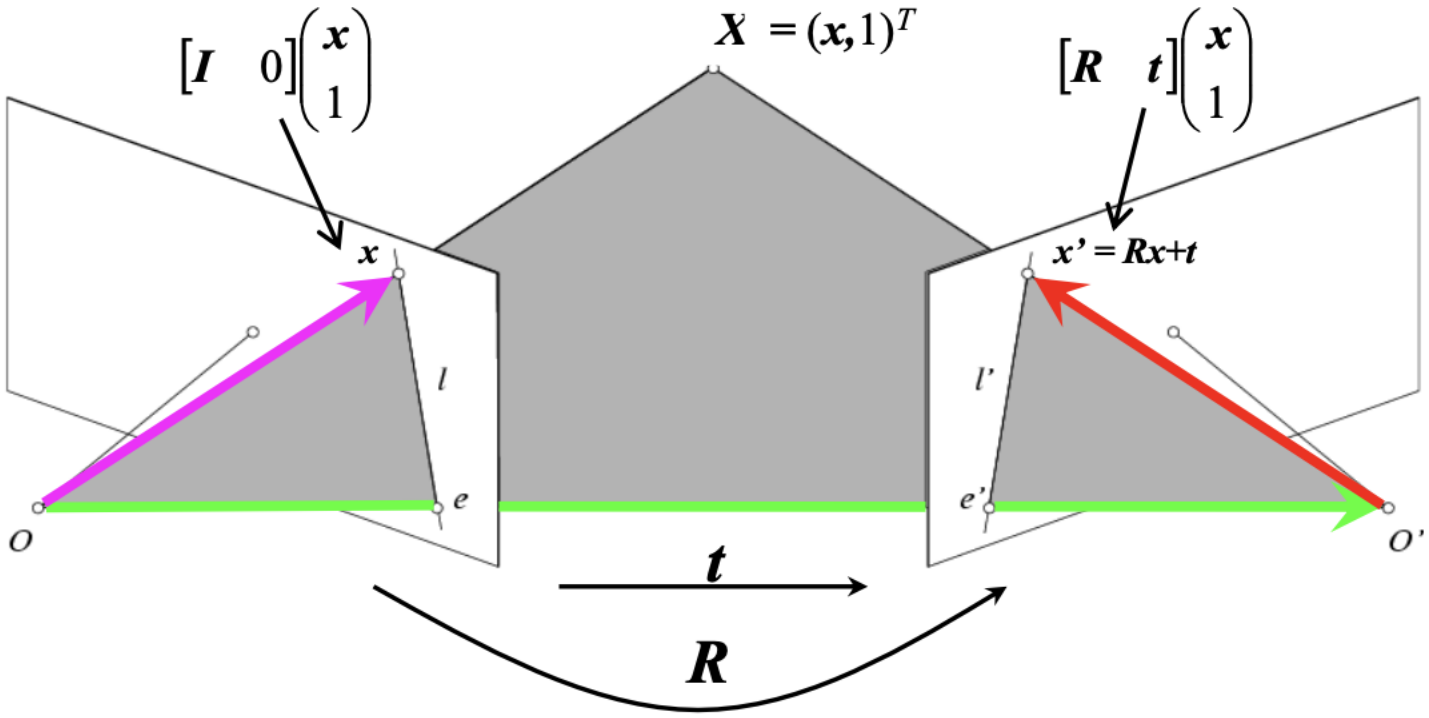


$$\hat{\mathbf{p}}^T \mathbf{E} \hat{\mathbf{p}}' = 0$$

$$\mathbf{E} = [\mathbf{t}]_\times \mathbf{R}$$

Essential Matrix

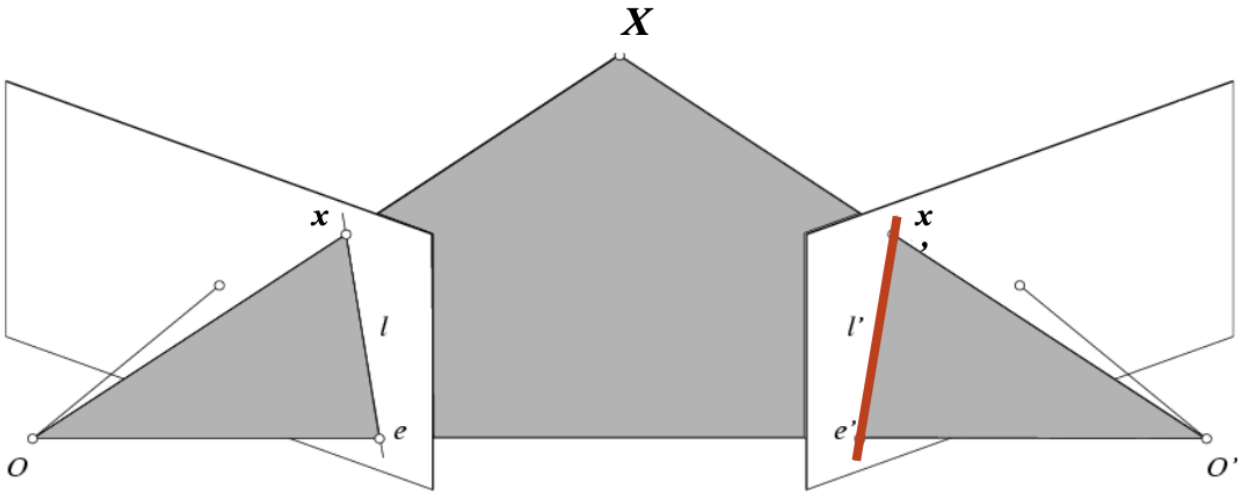
Epipolar constraint: Calibrated case



$$x' \cdot [t \times (Rx)] = 0 \quad \Rightarrow \quad x'^T [t_\times] R x = 0 \quad \Rightarrow \quad x'^T E x = 0$$

Essential Matrix
(Longuet-Higgins, 1981)

Epipolar constraint: Calibrated case



$$x'^T E x = 0$$

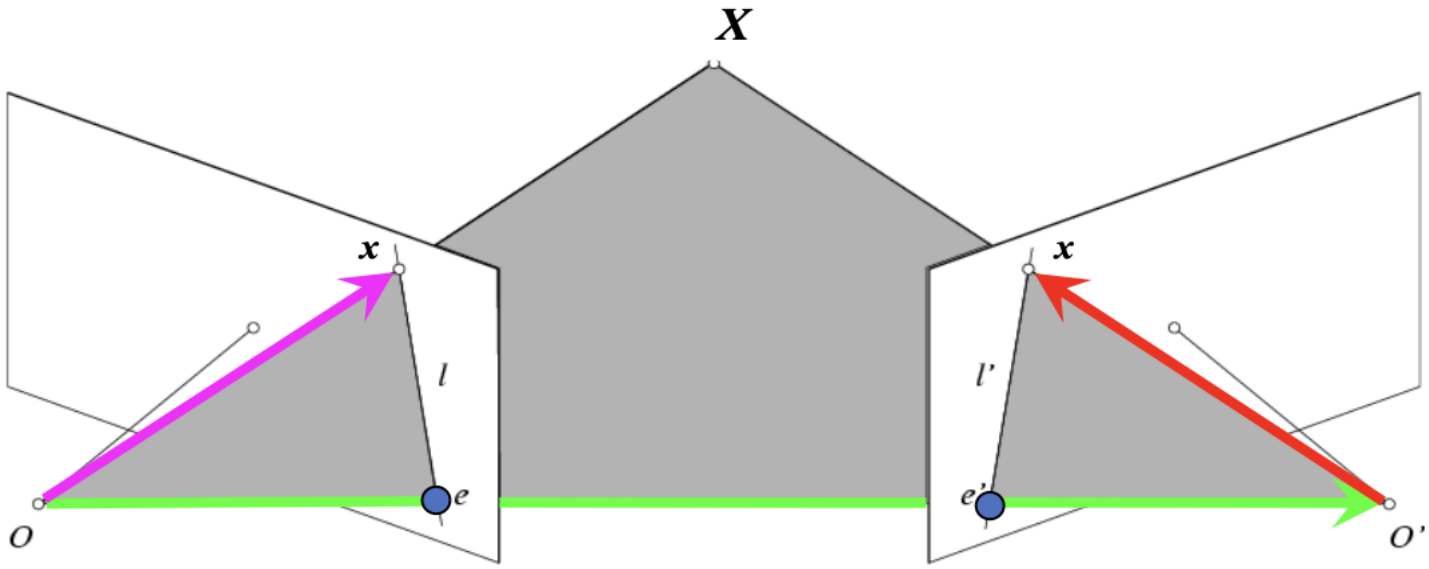
- Ex is the epipolar line associated with x' $(l' = Ex)$
- $E^T x'$ is the epipolar line associated with x $(l = E^T x')$
- $Ee = 0$ and $E^T e' = 0$
- E is **singular** (rank two)
- $E = [t_\times]R$ has five degrees of freedom

$$2T + 3R$$

- Recall: a line is given by $ax + by + c = 0$ or

$$\mathbf{l}^T\mathbf{x} = 0 \quad \text{where} \quad \mathbf{l} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

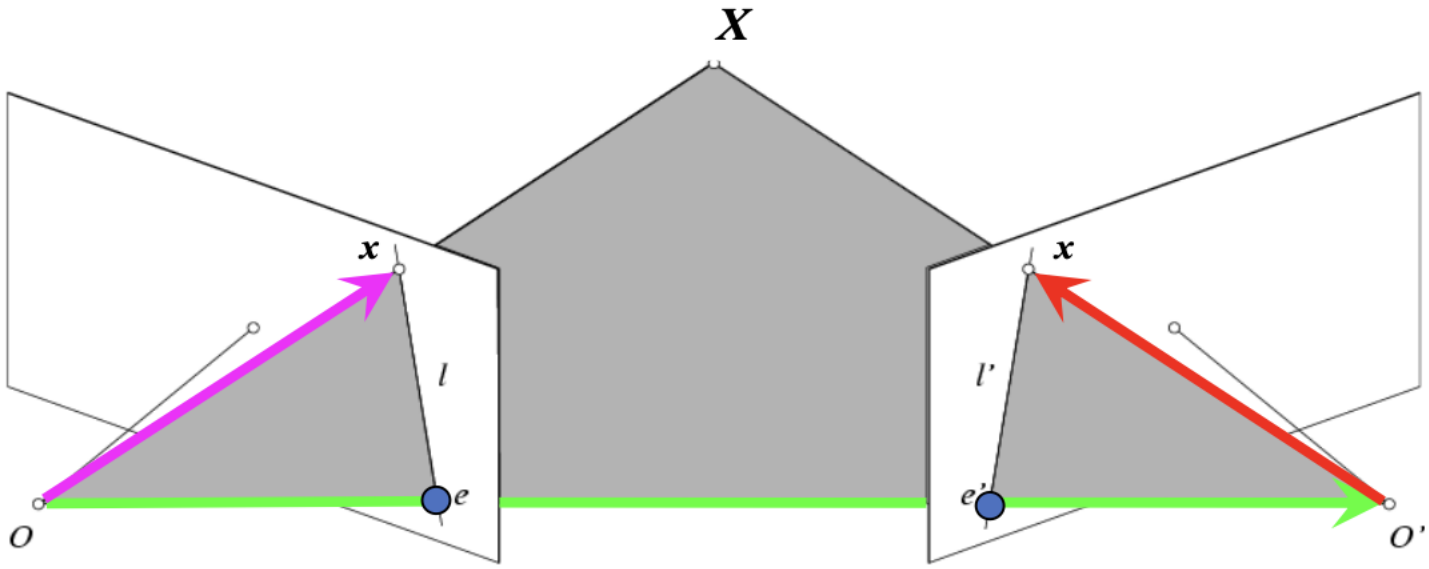
Epipolar constraint: Uncalibrated case



- The calibration matrices K and K' are unknown.
- We can write the epipolar constraint in terms of unknown normalized coordinates:

$$\hat{\boldsymbol{x}}'^{T}\boldsymbol{E}\hat{\boldsymbol{x}}=0 \qquad \hat{\boldsymbol{x}}=\boldsymbol{K}^{-1}\boldsymbol{x}, \quad \hat{\boldsymbol{x}}'=\boldsymbol{K}'^{-1}\boldsymbol{x}'$$

Epipolar constraint: Uncalibrated case



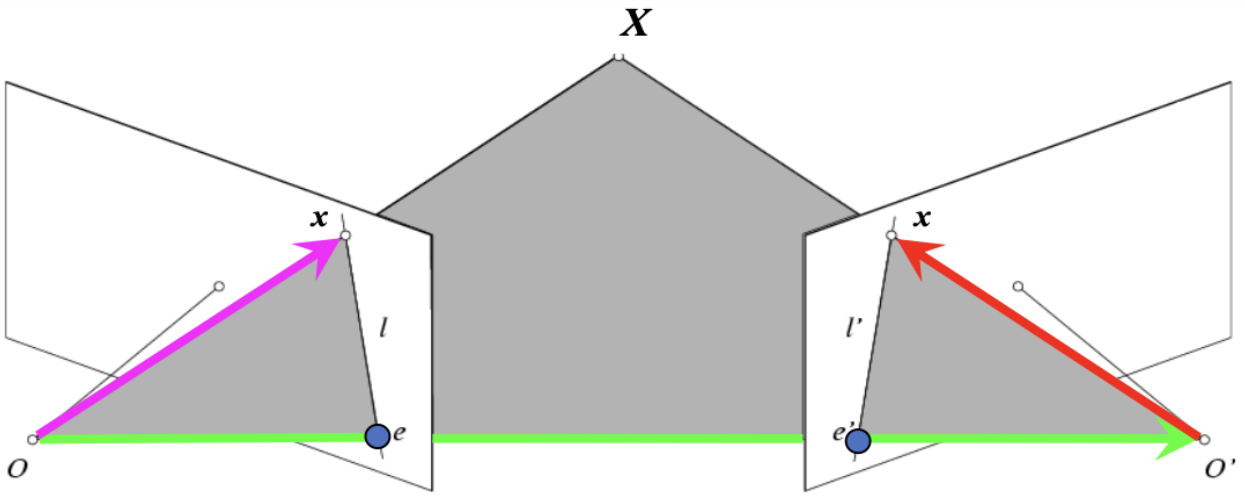
$$\hat{\boldsymbol{x}}'^{T}\boldsymbol{E}\hat{\boldsymbol{x}}=0 \quad \Rightarrow \quad \boldsymbol{x}'^{T}\boldsymbol{F}\boldsymbol{x}=0 \quad \text{with} \quad \boldsymbol{F}=\boldsymbol{K}'^{-T}\boldsymbol{E}\boldsymbol{K}^{-1}$$

$$\hat{\boldsymbol{x}}=\boldsymbol{K}^{-1}\boldsymbol{x}$$

$$\hat{\boldsymbol{x}}'=\boldsymbol{K}'^{-1}\boldsymbol{x}'$$

Fundamental Matrix
(Faugeras and Luong, 1992)

Epipolar constraint: Uncalibrated case



$$\hat{\mathbf{x}}'^T \mathbf{E} \hat{\mathbf{x}} = 0 \quad \Rightarrow \quad \mathbf{x}'^T \mathbf{F} \mathbf{x} = 0 \quad \text{with} \quad \mathbf{F} = \mathbf{K}'^{-T} \mathbf{E} \mathbf{K}^{-1}$$

- Fx is the epipolar line associated with x' $(l' = Fx)$
- $F^T x'$ is the epipolar line associated with x $(l = F^T x')$
- $Fe = 0$ and $F^T e' = 0$
- F is singular (rank two)
- F has seven degrees of freedom $(5E + 2K)$

Estimating the Fundamental Matrix : 8-point algorithm

- Solve a system of homogeneous linear equations
  - a. Write down the system of equations

$$\boldsymbol{x}^T F \boldsymbol{x}' = 0$$

$$uu'f_{11} + uv'f_{12} + uf_{13} + vu'f_{21} + vv'f_{22} + vf_{23} + u'f_{31} + v'f_{32} + f_{33} = 0$$

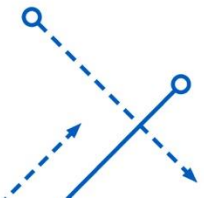
$$\mathrm{A}\boldsymbol{f} = \begin{bmatrix} u_1u_1' & u_1v_1' & u_1 & v_1u_1' & v_1v_1' & v_1 & u_1' & v_1' & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ u_nu_v' & u_nv_n' & u_n & v_nu_n' & v_nv_n' & v_n & u_n' & v_n' & 1 \end{bmatrix} \begin{bmatrix} f_{11} \\ f_{12} \\ f_{13} \\ f_{21} \\ \vdots \\ f_{33} \end{bmatrix} = \mathbf{0}$$

  - b. Solve f from $Af = 0$ using SVD (refer to Project 1).

```
[U, S, V] = svd(A);
f = V(:, end);
F = reshape(f, [3 3])';
```

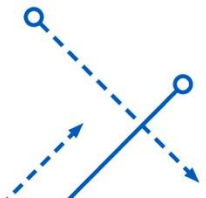
Q & A

- Why do we need 4 points for homography but 8 points for fundamental matrix calculation?
  - In the case of fundamental matrix, each point relates to only one constraint, while in homograph, each point is related to two constraints.
- Can we use 7 points solve fundamental matrix?
  - In fact, the fundamental matrix only has 7 degrees of freedom. In this case, the rank-2 constraint must be enforced during the computations.



Estimating the Fundamental Matrix

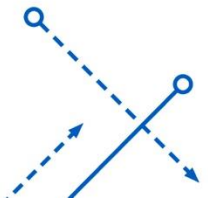
- 8-point algorithm
  - Least squares solution using SVD on equations from 8 pairs of correspondences.
- 7-point algorithm
  - least squares to solve for null space (two vectors) using SVD and 7 pairs of correspondences.
  - Solve for linear combination of null space vectors that satisfies $\det(F) = 0$
- Minimize reprojection error
  - Non-linear least squares
- Note: estimation of F (or E) is degenerate for a planar scene.



Correspondence Search

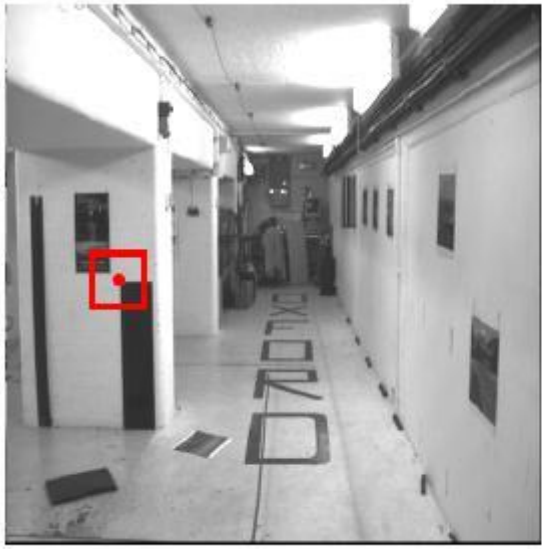
- Other “soft” constraints (To cover)
  - 1. Similarity
  - 2. Uniqueness
  - 3. Disparity gradient
  - 4. Ordering

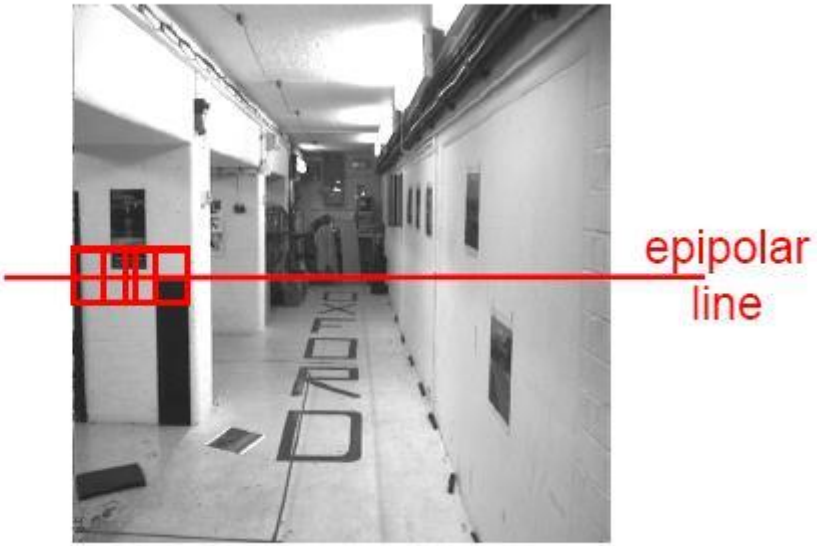
- To find matches in the image pair, we will assume
  - Most scene points visible from both views
  - Matched regions are similar in appearance



Dense correspondence search

- Neighborhoods of corresponding points are similar in intensity patterns.



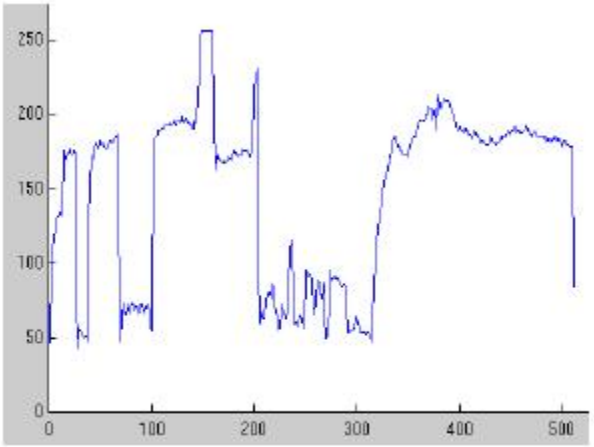


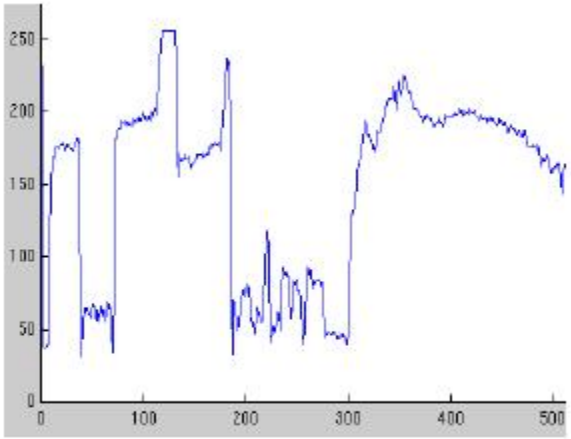
Intensity profiles





Intensity profiles



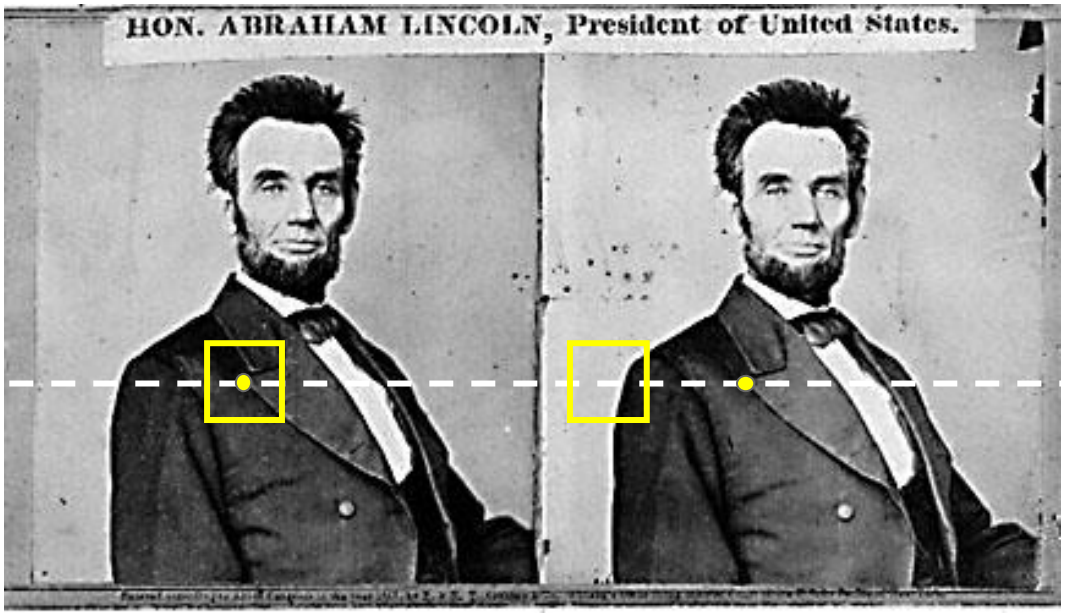


- Clear correspondence between intensities, but also noise and ambiguity

Source: Andrew Zisserman

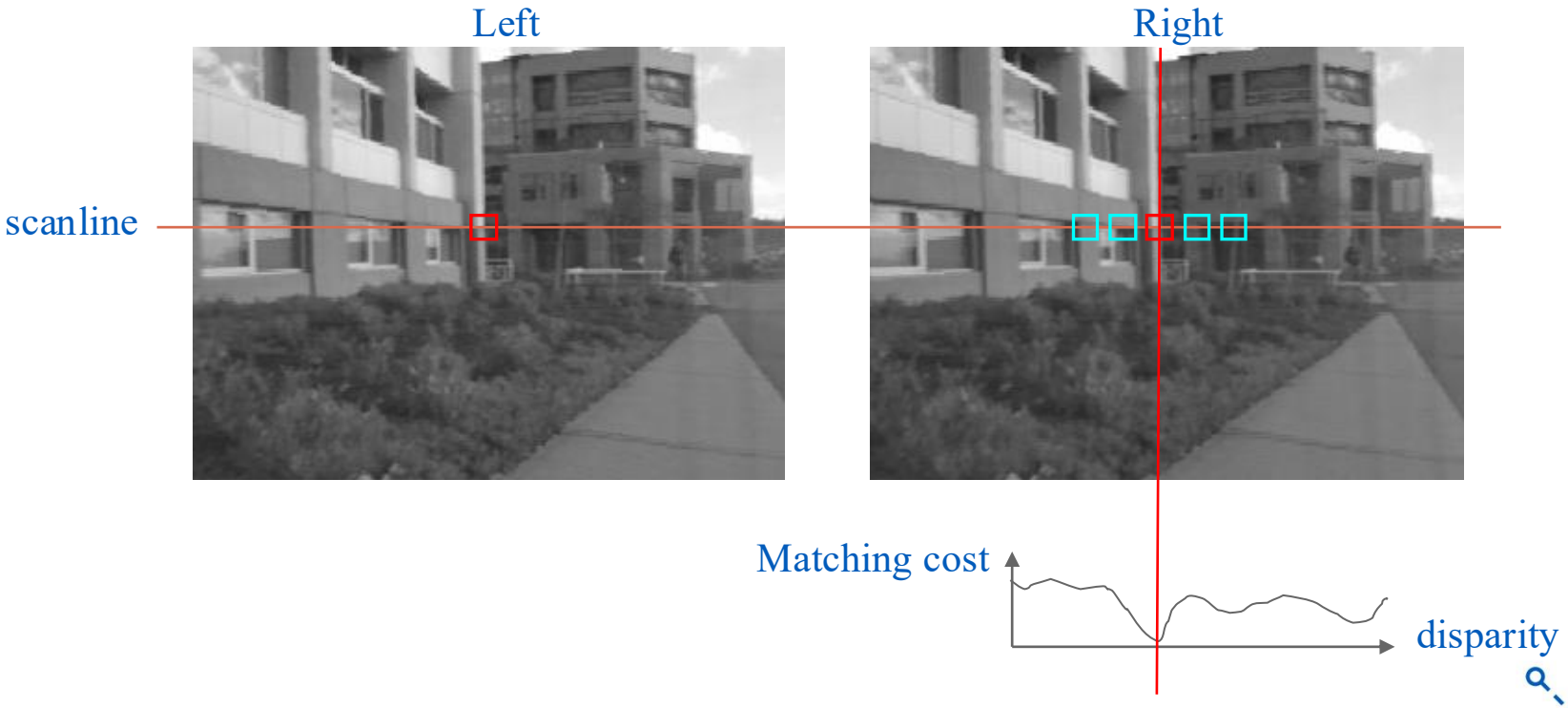
Dense correspondence search

- For each epipolar line
  - For each pixel / window in the left image
    - Compare with every pixel / window on same epipolar line
    - Pick position with minimum match cost
      - SSD, normalized correlation

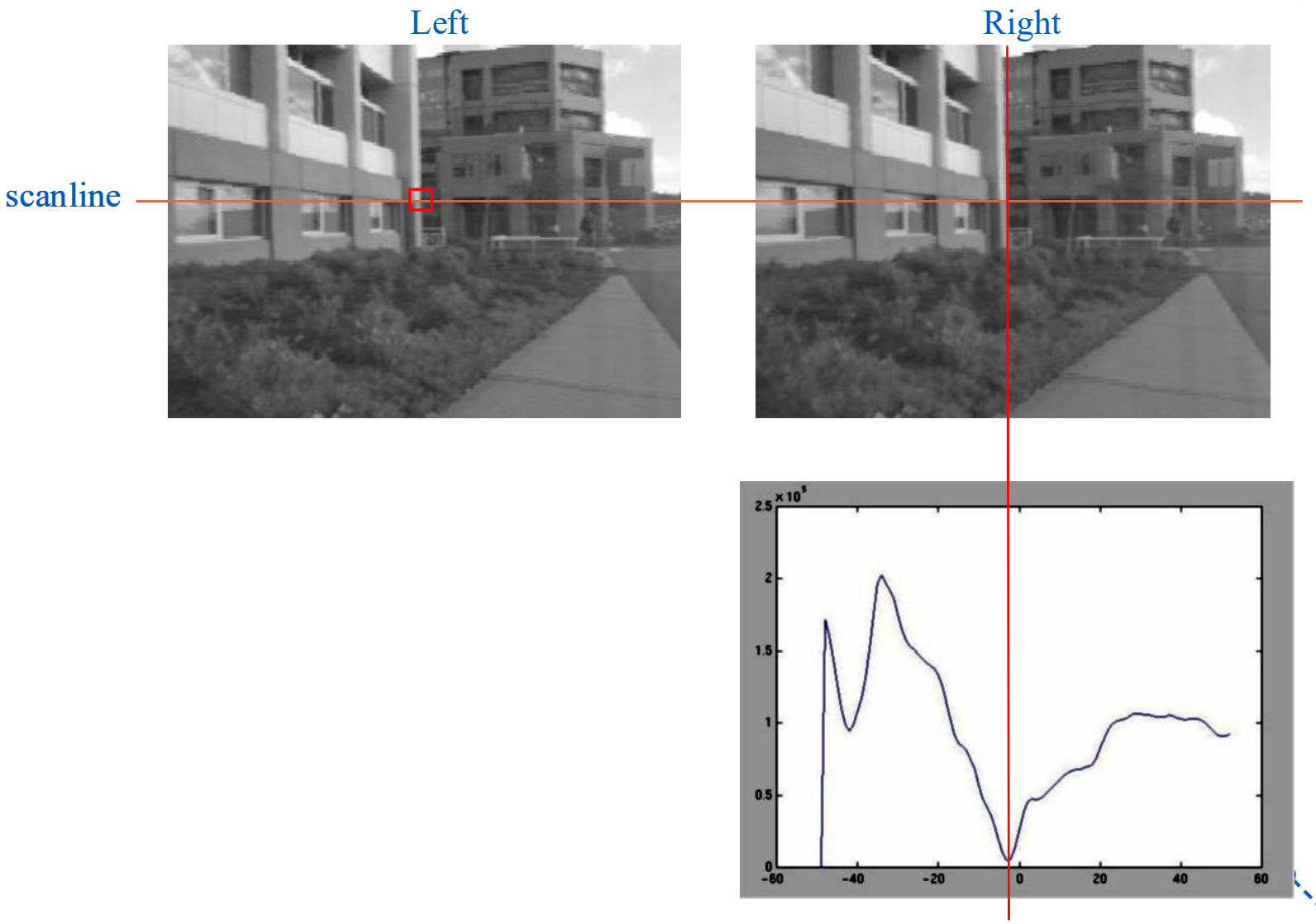


Similarity constraints

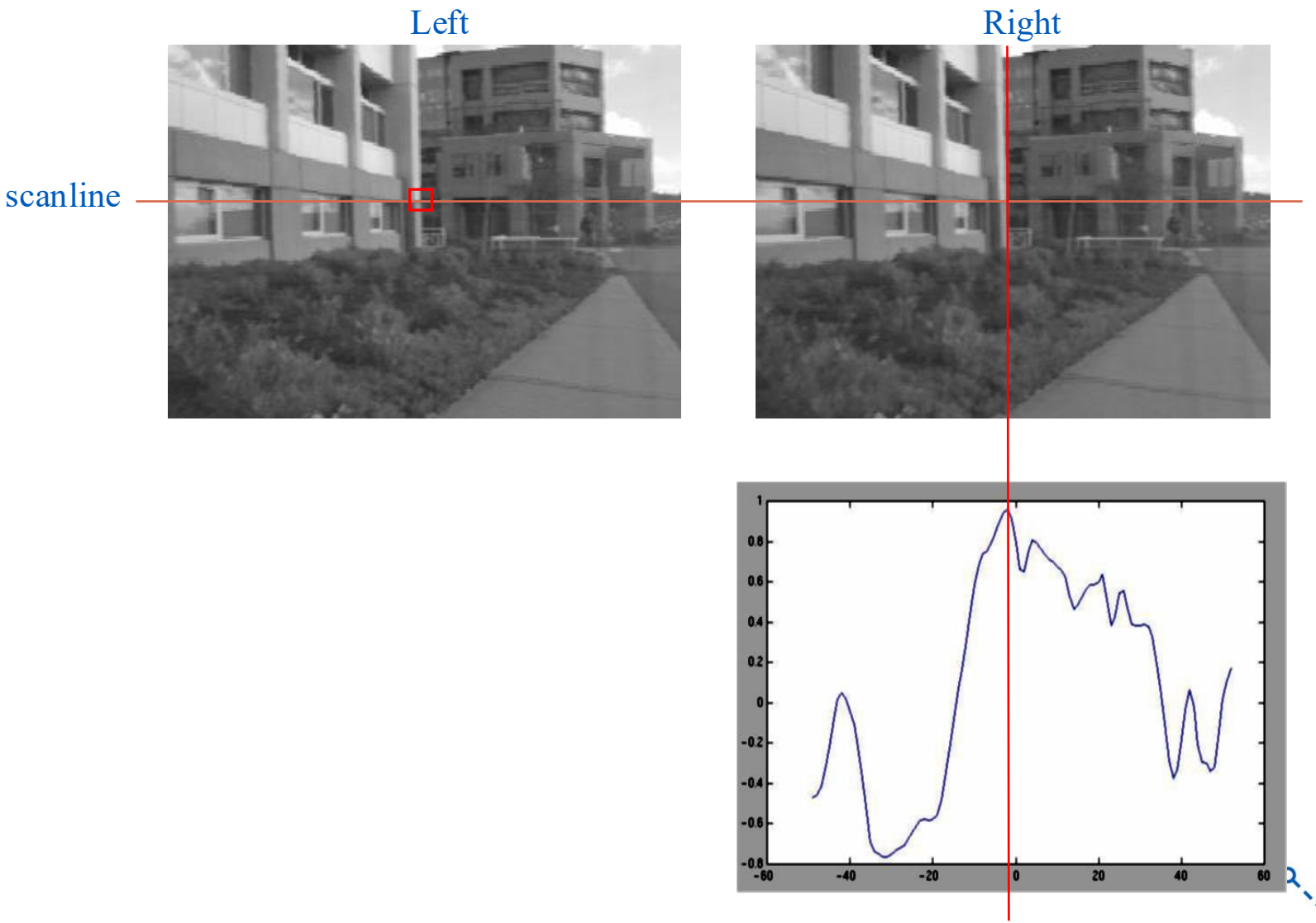
- Slide a window along the right scanline and compare contents of that window with the reference window in the left image
- Matching cost: SSD or normalized correlation



Similarity constraints: SSD



Similarity constraints: Norm. Corr.



Correlation-based window matching

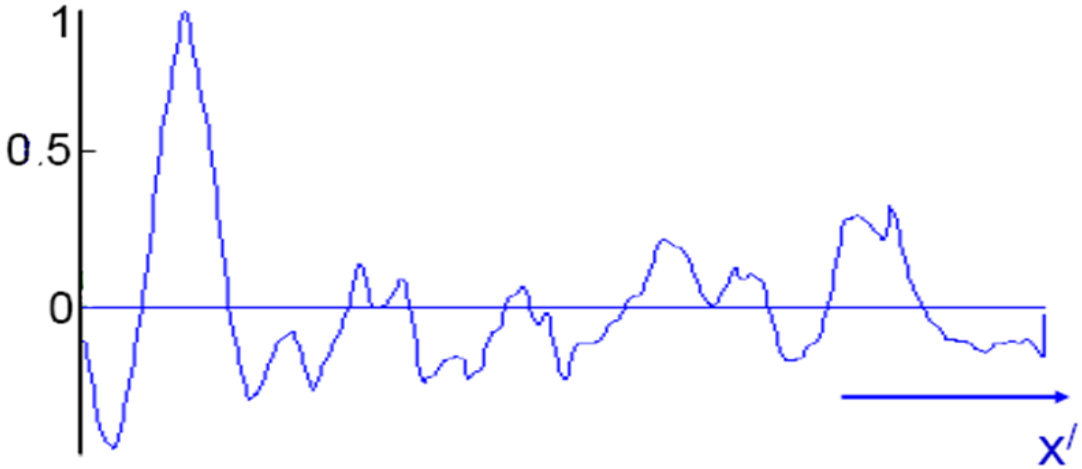




left image band (x)



right image band (x')



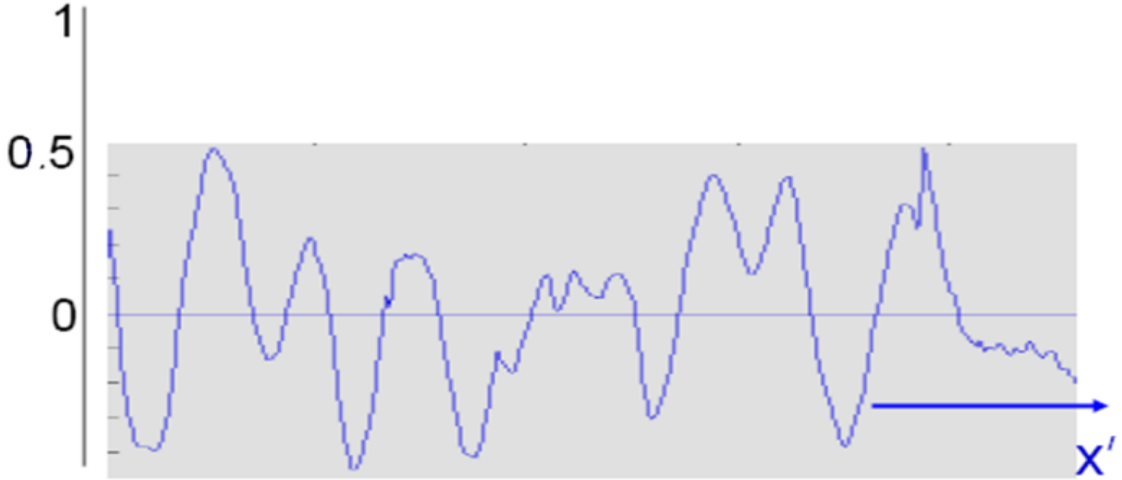
cross correlation

disparity = $x' - x$

Correlation-based window matching





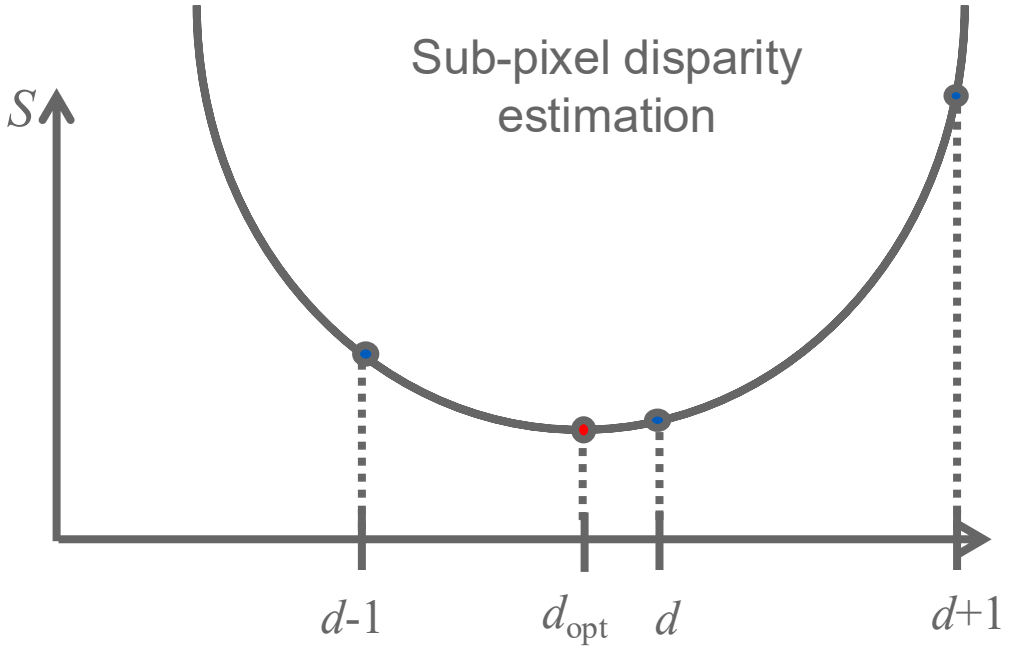


cross correlation

Textureless regions are non-distinct; high ambiguity for matches.

Sub-pixel disparity estimation

- Let S be the SSD

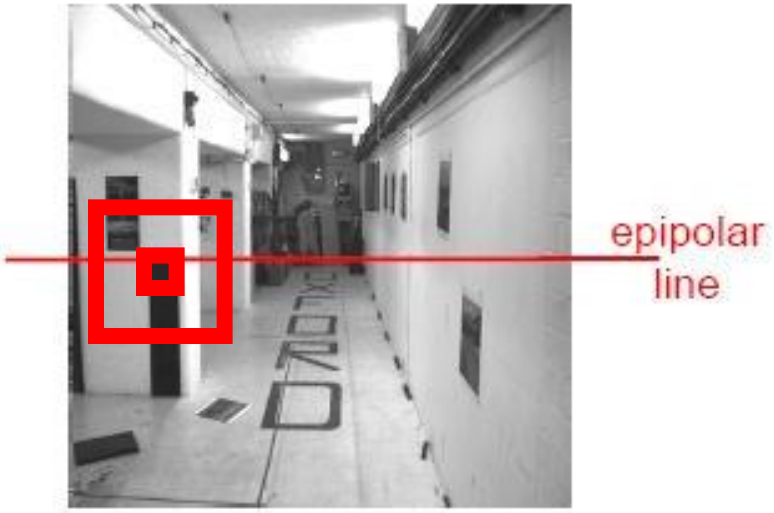


- $S(d) = ad^2 + bd + c$
  - $S(0) = c$
  - $S(1) = a + b + c$
  - $S(-1) = a - b + c$
- Solving this, we obtain:
  - $a = (S(1) + S(-1) - 2S(0))/2$
  - $b = (S(1) - S(-1))/2$
  - $c = S(0)$
- $S'(d) = 2ad + b = 0$

$$d_{opt} = \frac{(S(-1) - S(1))}{2(S(1) + S(-1) - 2S(0))}$$

Effect of window size





Source: Andrew Zisserman

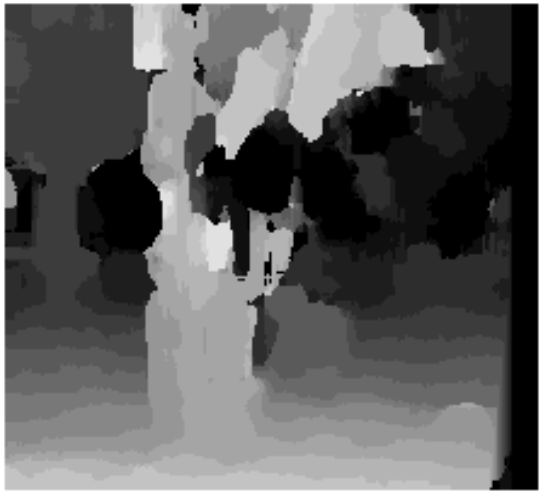
Effect of window size

- large enough to have sufficient intensity variation
- small enough to contain only pixels with about the same disparity.

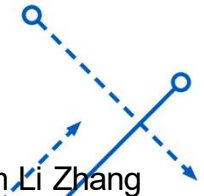




W = 3



W = 20



from Li Zhang

Uniqueness constraint

- Up to one match in right image for every point in left image

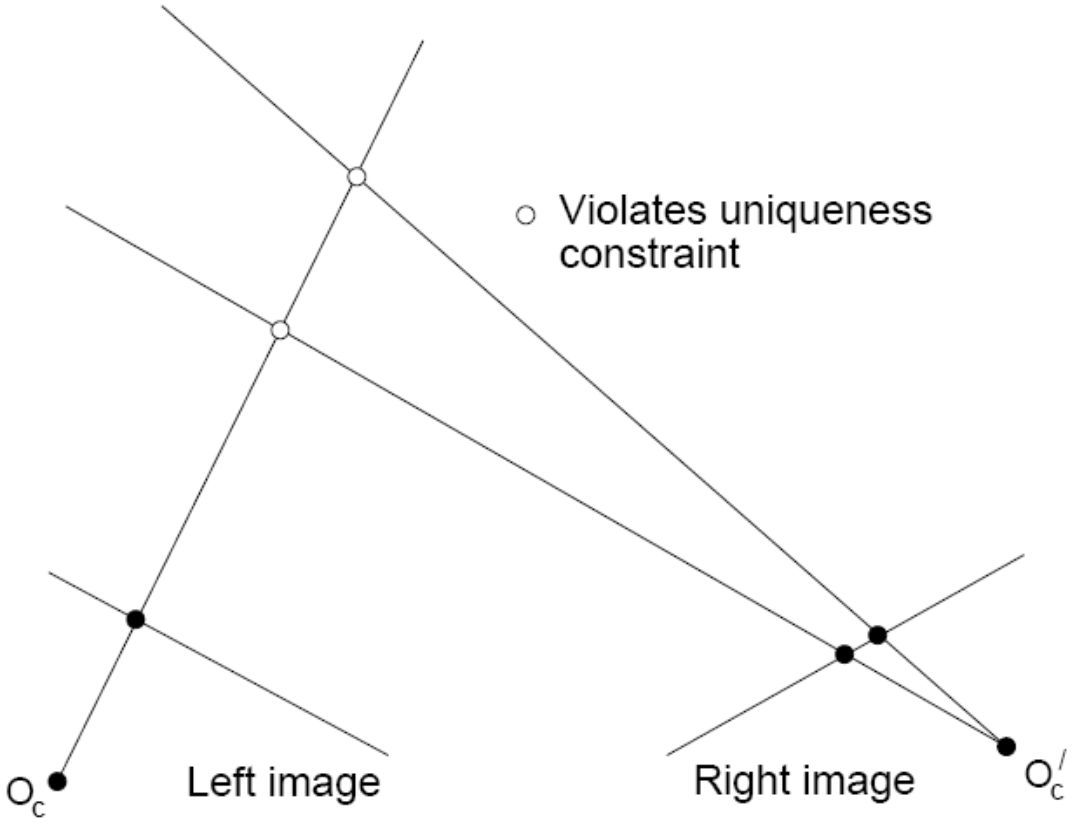
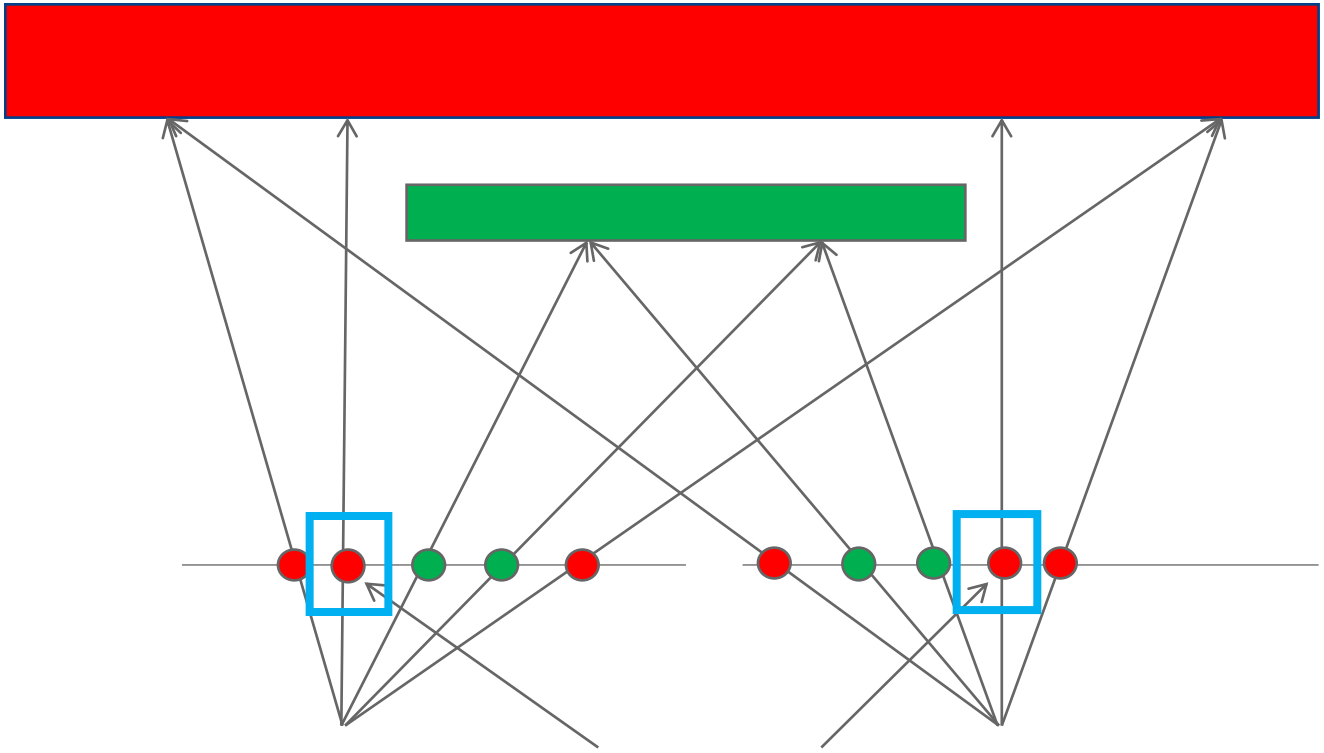


Figure from Gee & Cipolla 1999

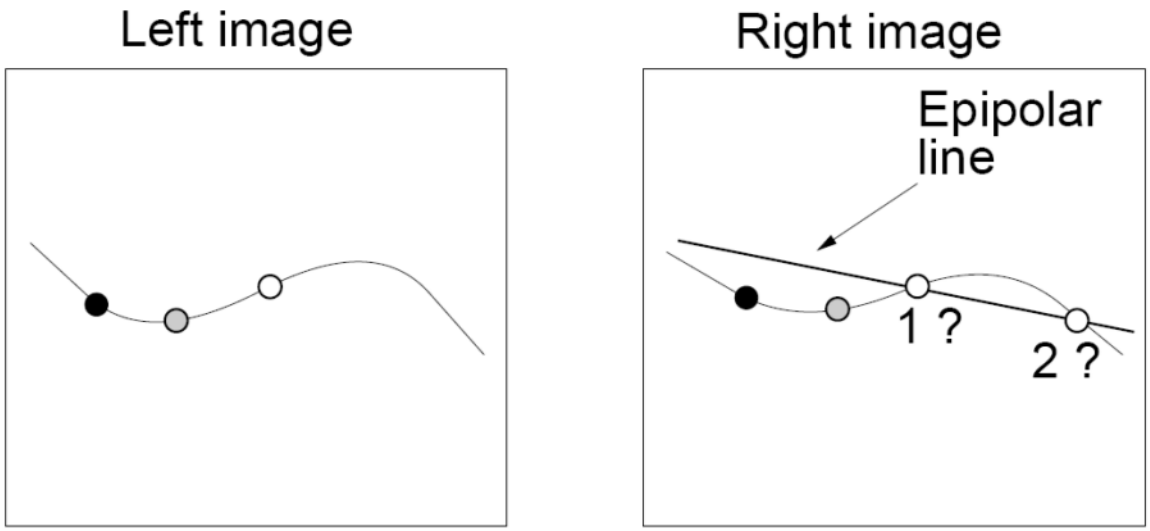
Problem: Occlusion

- Uniqueness says “up to one match” per pixel
- What if there is no match?



Disparity gradient constraint

- Assume piecewise continuous surface, so we want disparity estimates to be locally smooth



Given matches ● and ○, point ○ in the left image must match point 1 in the right image. Point 2 would exceed the disparity gradient limit.



Figure from Gee & Cipolla 1999

Ordering constraint

- Points on **same surface** (opaque object) will be in same order in both views

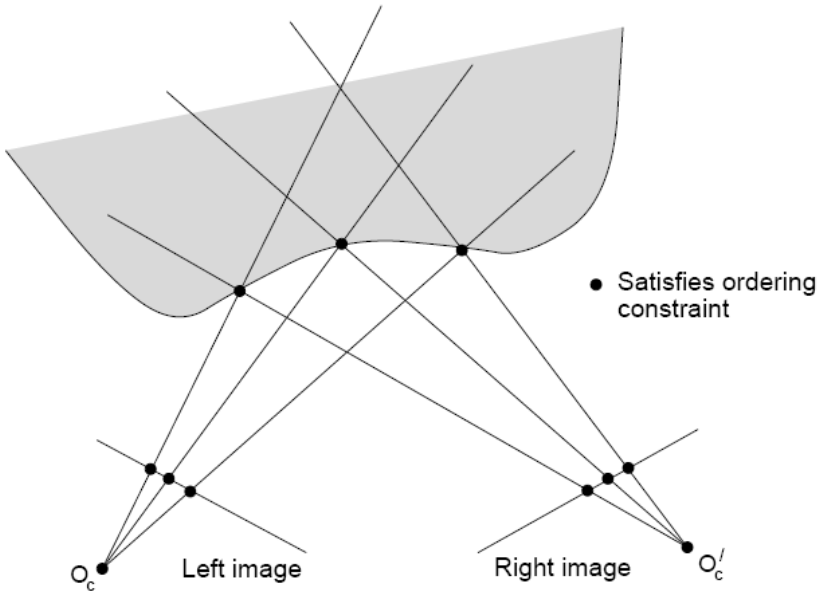
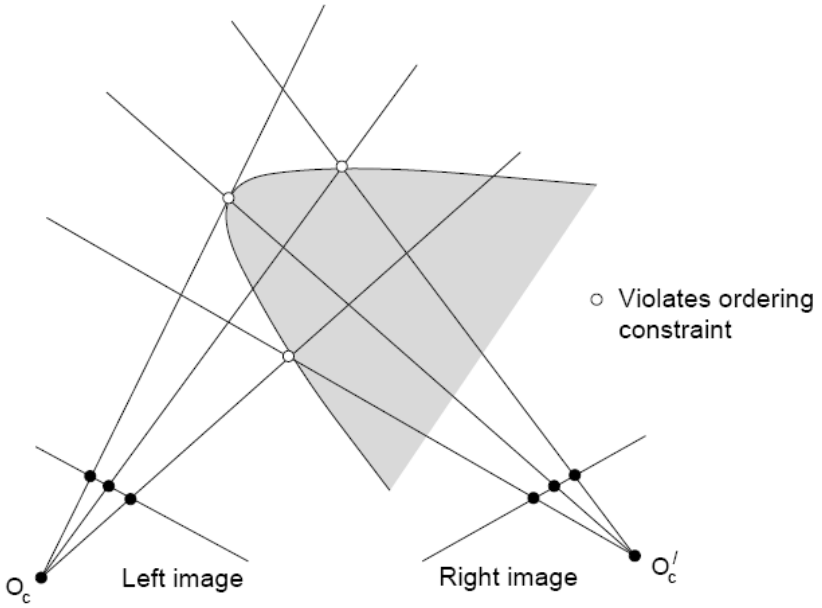
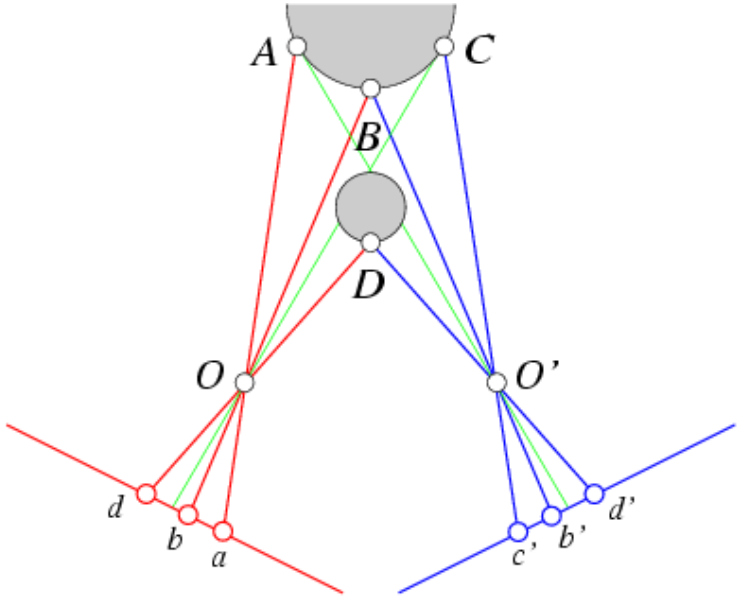


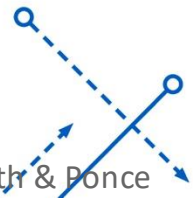
Figure from Gee & Cipolla 1999

Ordering constraint

- Won't always hold, e.g., consider transparent object, or an occluding surface

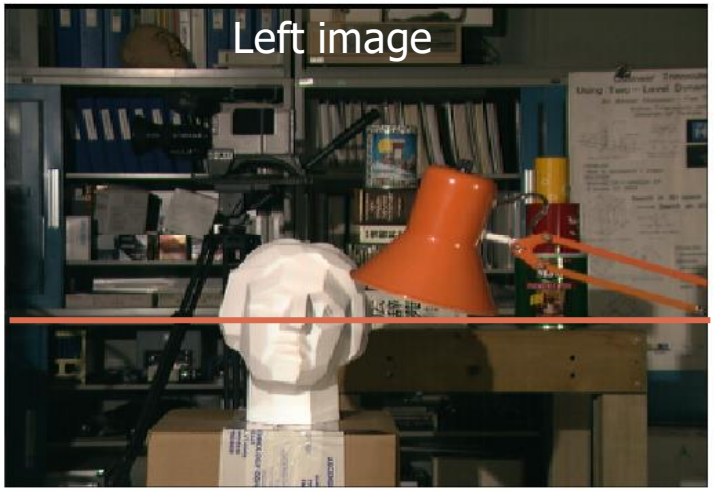


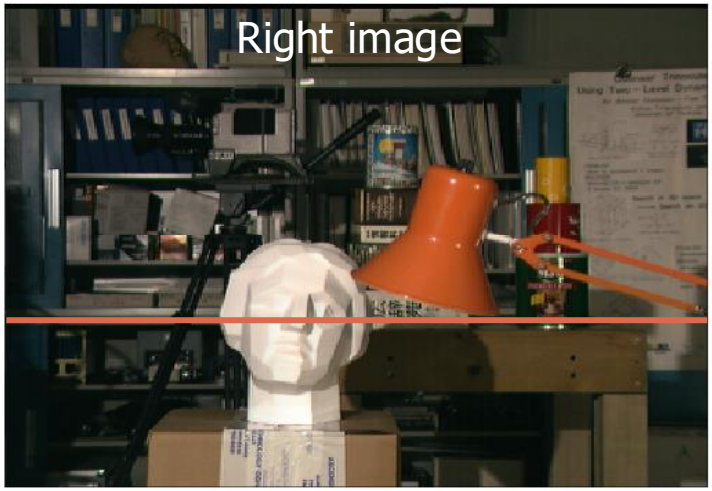


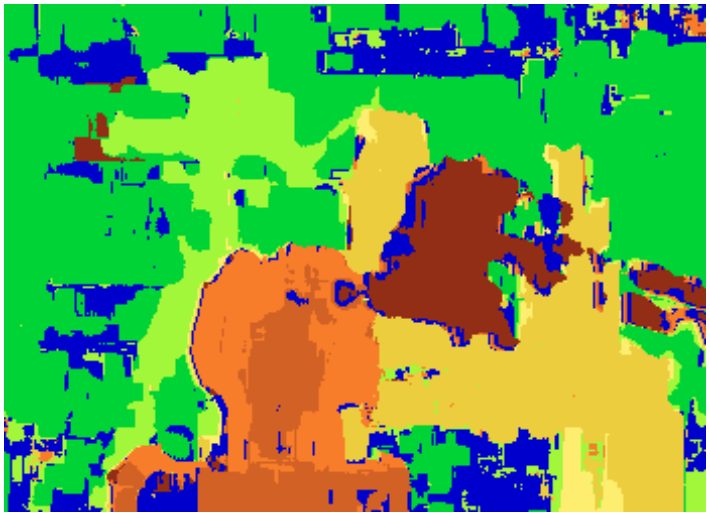


Figures from Forsyth & Ponce

Results with window search







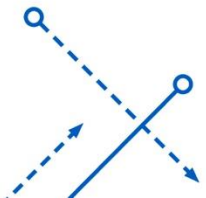
Window-based matching
(best window size)



Ground truth

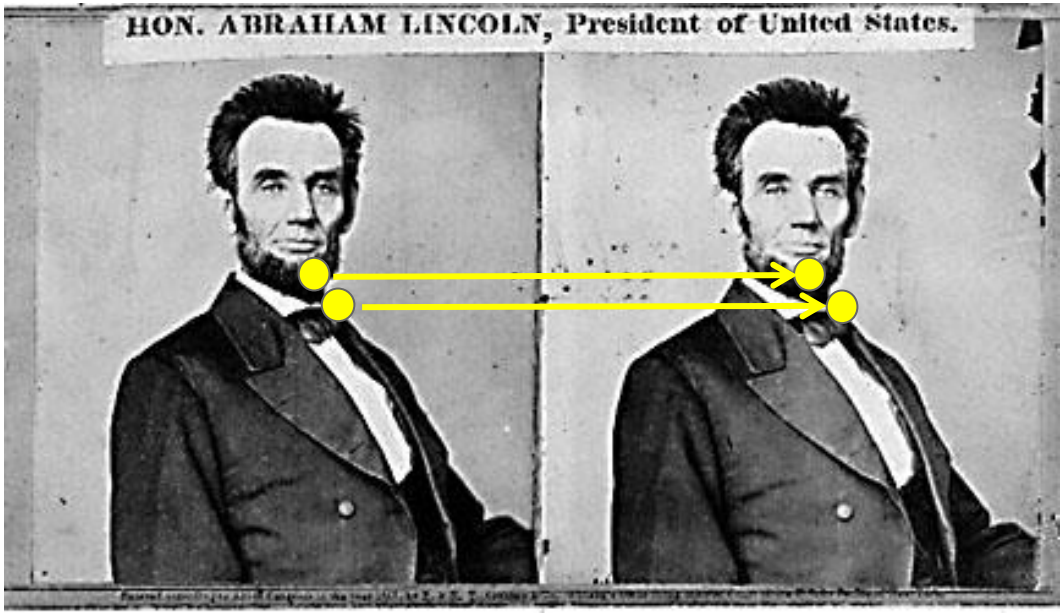
Better solutions

- Beyond individual correspondences estimation
- Optimize correspondence assignments jointly
  - Scanline at a time (DP)
  - Full 2D grid (graph cuts)



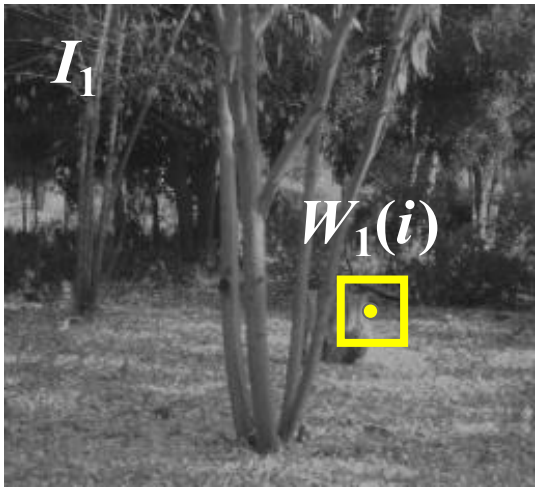
Stereo as energy minimization

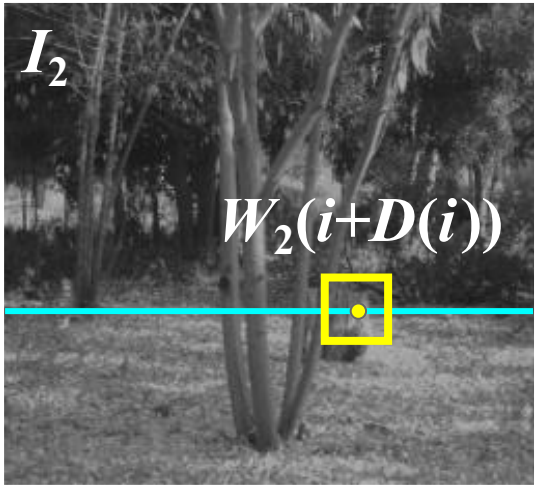
- What defines a good stereo correspondence?
  - Match quality
    - Want each pixel to find a good match in the other image
  - Smoothness
    - Adjacent pixels often move about the same amount.

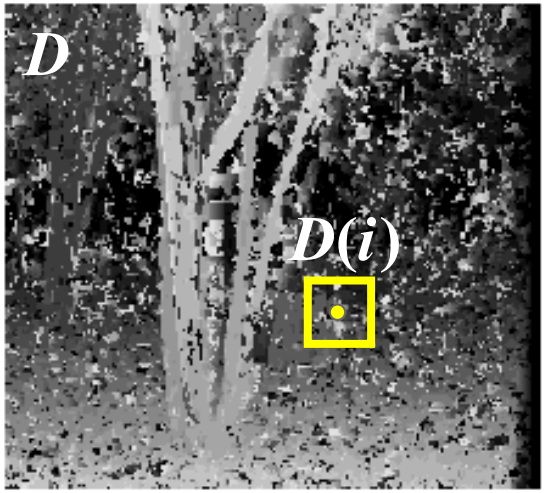


Stereo matching as energy minimization

- Energy functions of this form can be minimized using *graph cuts*







$$E = \alpha E_{\text{data}}(I_1, I_2, D) + \beta E_{\text{smooth}}(D)$$

$$E_{\text{data}} = \sum_i \left(W_1(i) - W_2(i + D(i))\right)^2$$

$$E_{\text{smooth}} = \sum_{\text{neighbors}\, i,j} \rho\left(D(i) - D(j)\right)$$

Y. Boykov, O. Veksler, and R. Zabih, Fast Approximate Energy Minimization via Graph Cuts, PAMI 2001

Better results…



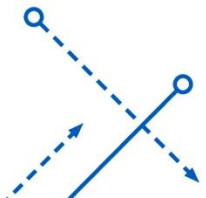
Graph cut method



Ground truth

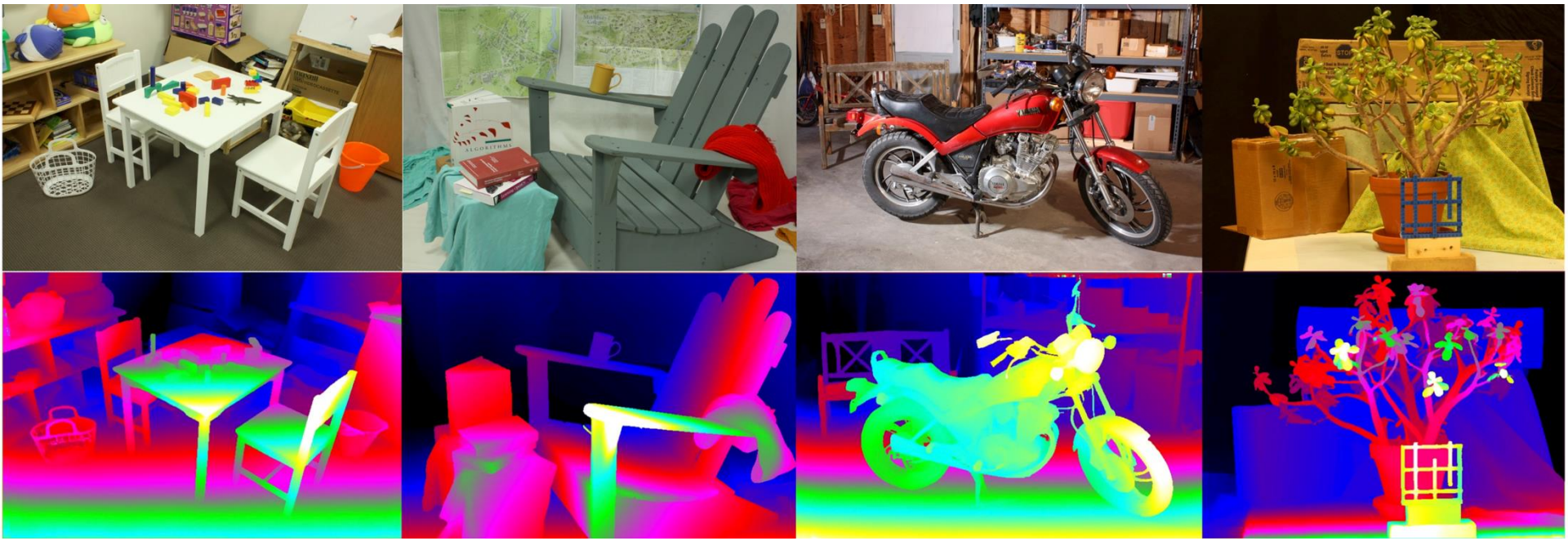
Challenges

- Low-contrast
  - Textureless image regions
- Occlusions
- Violations of brightness constancy
  - e.g., specular reflections
- Really large baselines
  - Foreshortening and appearance change
- Camera calibration errors



Data-driven Stereo Matching

- Data-driven estimated terms in energy minimization.
  - Disparity Proposal Network
  - Neural message passing



Tongfan Guan, Chen Wang, Yu-hui Liu, Neural Markov Random Field for Stereo Matching, IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024.

Contents

- Stereo Vision
  - Epipolar Geometry
    - Disparity
    - Epipolar line
    - Epipolar constraints
    - Rectification
    - Essential Matrix
    - Fundamental Matrix
  - Stereo Matching
    - Correspondence Search
    - Similarity
    - Uniqueness
    - Disparity gradient
    - Ordering

